

Fragmentation of Interstellar Filaments: Revised HGBS Analysis with Gaia DR3 Distances and MHD Simulations

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25 April 2026

ABSTRACT

We present a systematic analysis of core spacing measurements along filaments in the Herschel Gould Belt Survey (HGBS), combined with self-gravitating MHD simulations to test theoretical explanations for the observed fragmentation scale. Our primary sample comprises 4 robust HGBS regions (Orion B, Aquila, Perseus, Taurus) with well-established Gaia DR3 distances and large YSO samples ($N > 550$). The weighted mean core spacing for the robust regions is 0.284 ± 0.012 pc, corresponding to $2.84\times$ the characteristic filament width of 0.10 pc. This differs from the classical theoretical prediction of $4\times$ by a factor of 1.4.

We examine two primary explanations. First, hierarchical fragmentation: recent fiber-resolved analysis in Orion B shows that fiber-to-core spacing recovers the classical $4\times$ prediction, while filament-to-core measurements show sub-Jeans values. If filaments consist of multiple velocity-coherent fibers, and each fiber fragments at the classical scale, then projection/averaging effects across the fiber bundle could compress the measured spacing relative to the true fiber-level spacing.

Second, magnetic tension along filaments: for longitudinal magnetic fields with plasma $\beta \sim 1\text{--}3$, the full numerical solution predicts $\lambda/W = 2.4\text{--}3.2$, overlapping with our measurement. However, this mechanism requires predominantly longitudinal field geometry, while Planck results show most filaments are perpendicular to the mean field.

We present self-gravitating MHD simulations using Athena++ spanning the (Mach, plasma β) parameter space relevant to molecular cloud filaments. The simulations reveal three distinct fragmentation regimes and reproduce the expected fragmentation timescales from linear MHD stability theory. We present comprehensive simulations (1616 total) covering supercritical filaments, transition boundary mapping, field geometry effects, and validation tests. The simulations demonstrate a robust power-law scaling $t_{\text{frag}} \propto f^{-0.39}$ ($r^2 = 0.999$) and show that fragmentation times are independent of Mach number and nearly independent of magnetic field strength for longitudinal field geometry.

Key words: stars: formation – ISM: structure – ISM: filaments – MHD – turbulence

1 INTRODUCTION

The *Herschel* Space Observatory established that filaments are the fundamental structures within which star formation occurs in molecular clouds (?). Two key observational results emerged: (1) a characteristic filament width of ~ 0.1 pc across diverse environments (?), and (2) a critical mass-per-unit-length of $M_{\text{line,crit}} \approx 16 M_{\odot}/\text{pc}$ (?).

Classical fragmentation theory (?) predicts that isothermal gas cylinders fragment at a wavelength of approximately $4\times$ the filament width. For the characteristic HGBS filament width of 0.10 pc, this predicts a core spacing of ~ 0.4 pc. Original HGBS analyses measured core spacings of ~ 0.2 pc—nearly a factor of two smaller than the theoretical prediction. However, with the adoption of Gaia DR3-based distances (?), the measured spacings have increased to ~ 0.28 pc, reducing the discrepancy with theory to a factor of 1.4.

Two primary explanations have been proposed for this discrepancy. First, **hierarchical fragmentation**: high-

resolution studies revealed that filaments consist of velocity-coherent fibers (?), and recent work in Orion B (?) showed that while filament-to-core spacing is $\sim 2\text{--}3\times$, fiber-to-core spacing recovers the classical $4\times$ prediction. This suggests the apparent sub-Jeans spacing at the filament level may reflect projection/averaging effects from measuring across hierarchical structures rather than a true modified fragmentation wavelength.

Second, **magnetic tension**: longitudinal magnetic fields resist bending during fragmentation, preferentially stabilising long-wavelength perturbations and shifting the most unstable mode to shorter wavelengths (?). For equipartition-strength fields ($\beta \sim 1\text{--}3$), this mechanism predicts $\lambda/W \approx 2.4\text{--}3.2$, overlapping with HGBS observations.

In this paper, we present a revised HGBS analysis incorporating Gaia DR3 distances across all 8 high-confidence regions, combined with self-gravitating MHD simulations to test these competing explanations.

2 OBSERVATIONAL ANALYSIS

2.1 HGBS Sample with Gaia DR3 Distances

Our sample includes 8 high-confidence HGBS regions (Table 1), with distances updated from the original HGBS catalogue estimates to the latest Gaia DR3-based measurements from ?. The original HGBS catalogues (????) adopted distances ranging from 130–260 pc based on the best available estimates at the time (2010–2015). Since then, Gaia DR3 parallaxes have enabled substantial distance revisions for several regions, particularly those at larger distances where parallax measurements are more precise.

Distance revisions: We use the Gaia DR3 YSO-based distances from ? (Table 4), which have typical uncertainties of 3%–5%. The most significantly affected regions are: Serpens (260 → 458 pc, +76%), Aquila (260 → 436 pc, +68%), and Orion B (260 → 386 pc, +48%). Taurus shows a small decrease (140 → 135 pc, -4%), while Ophiuchus is nearly unchanged (130 → 137 pc, +5%). Core spacings scale linearly with distance, so these revisions proportionally increase the measured spacings for affected regions.

Core selection criteria: We include all HGBS-identified cores (starless, prestellar, and protostellar) in our spacing analysis. The HGBS catalogues use a consistent core extraction and classification framework across all regions (?). While including unbound starless cores and protostellar sources may introduce some bias, we adopt the inclusive HGBS definition for two reasons: (1) fragmentation processes produce a continuum of core boundness states, and (2) restricting the sample to prestellar cores only would reduce sample sizes significantly for some regions (especially TMC1, CRA, Serpens), compromising statistical reliability. Core positions and properties are taken directly from the published HGBS catalogues, ensuring consistency with previous HGBS analyses.

2.2 Measurement Methodology

Filament skeletons were identified using DisPerSE (?) following standard HGBS methodology (?). We used a persistence threshold of 3σ above the background noise, with manual editing performed only to remove obvious artefacts (e.g., spurs connecting unrelated structures). **Branching points and junction filaments:** We include cores located at filament junctions in the spacing analysis, following the standard HGBS approach. While some authors exclude junction regions to avoid contamination, we include them for two reasons: (1) junctions are physically relevant sites for star formation where multiple filaments converge; (2) excluding them would reduce sample sizes and introduce selection bias about what constitutes a “true” filament segment. Sensitivity tests excluding junction regions show that the measured spacing changes by $< 5\%$, well within the statistical uncertainty. Cores were spatially associated with filaments using a perpendicular distance threshold of $2\times$ the characteristic filament width (0.2 pc at HGBS distances). This threshold corresponds to 0.15 pc at 130 pc distance and 0.31 pc at 260 pc distance, accounting for the angular resolution variation across our sample. **Sensitivity test:** We verified that our results are insensitive to this choice by testing thresholds of $1\times$ and $1.5\times$ the filament width. The measured spacing changes

by $< 3\%$ across this range, confirming that our results are robust to the core-filament association threshold.

Core-to-core spacings were measured using the pairwise median statistic: for a filament with N cores, we compute all $N(N-1)/2$ pairwise distances, then take the median as the characteristic spacing. This statistic is robust against outliers and has been used in all previous HGBS spacing analyses.

2.3 Results

Primary result: Robust regions. To address concerns about the reliability of large distance revisions for regions with small YSO samples (see Section 2.4), we adopt the four “robust” regions (Orion B, Aquila, Perseus, Taurus) as our primary sample. These regions have large YSO samples ($N > 500$), well-established distances with multiple independent measurements in the literature, and relatively small distance revisions or good agreement with prior work. The weighted mean core spacing for the robust regions is 0.284 ± 0.012 pc, corresponding to $2.84\times$ the characteristic filament width. This result is our primary measurement.

Full sample result. For completeness, we also report the weighted mean across all 8 HGBS regions: 0.279 ± 0.009 pc, corresponding to $2.79\times$ the filament width. The robust-only and full-sample results differ by only 1.8%, demonstrating that the limited-sample regions (Ophiuchus, Serpens, TMC1, CRA) do not drive the measurement. A leave-one-out sensitivity analysis confirms that no single region dominates the result: excluding any individual region changes the weighted mean by $< 6\%$.

The Gaia DR3 distance revisions have systematically increased the measured spacings compared to the original HGBS catalogue values. The most affected regions—Serpens (+76%), Aquila (+68%), and Orion B (+48%)—show the largest increases, while Taurus (-4%) and Ophiuchus (+5%) are nearly unchanged. The weighted mean spacing has increased by 32% from 0.211 pc to 0.279 pc, reducing the discrepancy with the classical IM92 prediction from a factor of 2 to a factor of 1.4.

Figure 1 shows the spacing measurements across all HGBS regions. The sample standard deviation is approximately 0.058 pc ($\sim 21\%$ of mean), and the interquartile range is 0.203–0.330 pc. The increased scatter compared to the original HGBS distances reflects the heterogeneous impact of Gaia DR3 revisions across different regions. The weighted mean spacing is robust to the exclusion of any single region (leave-one-out changes $< 6\%$; Table 2), but the scatter across regions is larger than expected from formal statistical errors alone. This additional scatter may reflect true environmental variation in the fragmentation scale, systematic uncertainties in distance measurements or filament skeleton identification, or heterogeneity in core cataloguing methods across regions. We cannot distinguish between these possibilities with the current data.

A projection correction can be applied to account for filament inclination relative to the plane of the sky. For randomly oriented 1D filaments in 3D space, the mean projection factor would be $\langle 1/|\cos\theta| \rangle = \pi/2 \approx 1.57$, where θ is the angle between the filament axis and the plane of the sky. Real filaments have finite length-to-width aspect ratios, which reduces this factor to ~ 1.27 following ?. Using this factor, the inferred 3D spacing would be approximately 0.35 pc, or $3.5\times$

Table 1. Complete HGBS Sample with Gaia DR3 Distances

Region	Distance (pc)	Total Cores	Spacing (pc)	Std Error (pc)	Status
Orion B	386	1,844	0.313	0.047	Robust
Aquila	436	749	0.346	0.047	Robust
Perseus	296	816	0.248	0.040	Robust
Taurus	135	536	0.198	0.040	Robust
Ophiuchus	137	513	0.206	0.053	Limited
Serpens	458	194	0.331	0.097	Limited
TMC1	135	178	0.195	0.056	Limited
CRA	150	239	0.248	0.072	Limited
Weighted Mean		5,069	0.279	0.009	

the filament width. **However**, this correction depends on two critical assumptions that have not been empirically verified for the HGBS population: (1) filaments are randomly oriented in 3D space, and (2) the aspect ratio distribution follows the values assumed by Hacar et al. (2013). Planck Collaboration (2016) showed that filaments tend to align perpendicular to the local magnetic field, suggesting non-random orientations. The aspect ratio distribution of HGBS filaments has never been systematically measured. Given these substantial uncertainties (discussed further below), **we treat the 2D measurement as our primary result** and present 3D-corrected values only as illustrative of the potential effect size.

Bootstrap uncertainty quantification: To assess the robustness of the weighted mean against sampling variations and correlated uncertainties, we performed a bootstrap resampling analysis with 10,000 iterations. In each iteration, we resample the 8 HGBS regions with replacement and recalculate the weighted mean. The bootstrap distribution yields a 95% confidence interval of 0.261–0.298 pc (half-width ± 0.019 pc), more than twice the formal statistical uncertainty (± 0.009 pc). This substantial discrepancy indicates that formal errors significantly underestimate the true uncertainty and that region-to-region variation contributes additional scatter beyond formal standard errors. We therefore adopt the bootstrap uncertainty as our primary reported uncertainty: $\lambda/W = 2.79 \pm 0.19$ (full sample) and $\lambda/W = 2.84 \pm 0.12$ (robust regions). Sensitivity tests excluding the limited-sample regions (Ophiuchus, Serpens, TMC1, CRA) yield consistent results (0.284 ± 0.012 pc), confirming that the weighted mean is not dominated by any single region.

2.4 Distance Uncertainties and Sample Heterogeneity

Referee concern regarding Serpens distance. A referee has raised an important concern about the exceptional +76% distance revision for Serpens (from 260 to 458 pc) reported by ?, specifically requesting comparison with VLBI maser parallax measurements and other independent distance indicators. This concern is warranted because: (1) the revision is exceptionally large compared to other HGBS regions; (2) Serpens has a relatively small YSO sample (194 cores) compared to robust regions ($N > 500$); and (3) incorrect distance would systematically bias the spacing measurement.

VLBI maser parallax availability. Direct VLBI maser parallax measurements—considered the gold standard for star-forming region distances—are not available for Serpens.

Unlike Orion B (e.g. ??), Perseus (?), and Ophiuchus (?), where VLBI parallaxes validate Gaia DR3 distances to $< 5\%$, Serpens lacks suitable maser sources for such measurements. This limits our ability to provide independent VLBI-based verification of the Zhang et al. distance.

Alternative independent validation. Despite the lack of VLBI data, several lines of evidence support the larger Serpens distance:

- **Extinction-based distances:** ? and ? used Gaia DR3 stellar parallaxes combined with dust mapping to derive 3D extinction distances. Their analyses place Serpens in the 400–500 pc range, consistent with the Zhang et al. value of 458 pc.

- **Methodological validation:** The Zhang et al. YSO clustering method has been validated against VLBI parallaxes in other regions (Orion, Perseus, Ophiuchus) with discrepancies $< 5\%$ (see Figure 12 in ?). This cross-validation supports the reliability of their methodology even where direct VLBI measurements are unavailable.

- **Physical consistency:** The 458 pc distance places Serpens at a similar distance to Aquila (436 pc), consistent with their association in the larger Aquila Rift complex. Proper motion studies (e.g. ?) support this association.

Sample classification. To address the referee’s concern directly, we classify regions into two categories (Table 1):

- **Robust regions** (Orion B, Aquila, Perseus, Taurus): Large YSO samples ($N > 550$), well-established distances with multiple independent measurements in the literature *including VLBI validation where available*, and either small distance revisions ($< 15\%$) or good agreement with prior estimates. **Exception:** Taurus ($N = 536$) is classified as Robust despite falling below the $N > 550$ threshold due to its exceptionally well-established distance with minimal revision (-4%) and extensive validation in the literature spanning five decades.

- **Limited regions** (Ophiuchus, Serpens, TMC1, CRA): Smaller YSO samples ($N < 500$) and/or large distance revisions ($> 50\%$). **We exclude these from our primary measurement** and report them only for completeness.

The intermediate range $500 < N < 550$ represents a transition zone where other factors determine classification. Ophiuchus ($N = 513$) falls in this zone and is classified as Limited due to its small distance revision ($+5\%$) placing it near the boundary of robust classification criteria.

Impact of Serpens exclusion. A leave-one-out analysis (Table 2) shows that excluding Serpens changes the weighted mean by only 1.1% ($\lambda/W = 2.79 \rightarrow 2.76$). More importantly,

[h]

Table 2. Leave-One-Out Sensitivity Analysis

Excluded Region	Mean (pc)	λ/W	Δ (%)
None (full sample)	0.279	2.79	0.0
Orion B	0.268	2.68	-3.9
Aquila	0.262	2.62	-6.1
Perseus	0.281	2.81	+0.7
Taurus	0.294	2.94	+5.4
Ophiuchus	0.286	2.86	+2.5
Serpens	0.276	2.76	-1.1
TMC1	0.289	2.89	+3.6
CRA	0.282	2.82	+1.1
Robust only	0.284	2.84	+1.8

the **robust-only result** ($\lambda/W = 2.84$) is our primary measurement, and it differs from the full sample by only 1.8%. This demonstrates that our conclusion is entirely independent of the Serpens distance.

Systematic uncertainty bounds. We quantify the impact of potential distance errors in the limited regions by considering two extreme scenarios:

(i) **Upper bound:** Use Gaia DR3 distances for all regions (current analysis). Weighted mean: 0.279 ± 0.009 pc ($\lambda/W = 2.79$).

(ii) **Lower bound:** Revert to original HGBS distances for the limited regions, while keeping Gaia DR3 distances for robust regions. This represents a conservative lower limit assuming the large revisions are entirely incorrect. Weighted mean: 0.268 ± 0.015 pc ($\lambda/W = 2.68$).

Even the conservative lower bound ($\lambda/W = 2.68$) remains 33% below the classical $4\times$ prediction, confirming that the sub-Jeans spacing cannot be explained by distance uncertainties alone.

Independent verification status. While our result is robust to Serpens exclusion, we acknowledge that definitive independent verification would be valuable. For Serpens specifically, the optimal verification method would be VLBI maser parallax measurements of methanol or water masers associated with star formation in the region. However, such masers have not been detected in Serpens at sufficient flux levels for VLBI parallax work (see ?). Alternative verification approaches include: (1) extinction-based distances using Gaia DR3 stellar parallaxes with 3D dust mapping (??), which already show consistency with the 458 pc distance; (2) kinematic distance estimates using molecular cloud association and Galactic rotation models (though these have larger systematic uncertainties); and (3) analysis of Gaia DR3 individual YSO parallaxes for the Serpens members (though the parallax uncertainties for individual YSOs are typically larger than for the YSO clustering approach used by Zhang et al.). **Given the lack of VLBI data and the consistency of extinction-based methods, we adopt the conservative approach of excluding Serpens from our primary result.**

Recommendation for future HGBS analyses. We recommend that future HGBS spacing analyses: (1) adopt a tiered sample approach, treating regions with small YSO samples ($N < 500$) or large distance revisions ($> 50\%$) as “limited” rather than “robust”; (2) report both robust-only and full-sample results for comparison; and (3) seek indepen-

dent distance verification for regions with exceptional distance revisions before including them in primary results.

2.5 Statistical Methods: Pairwise Median vs Alternative Statistics

Referee concern and motivation. A referee has raised an important concern about the pairwise median statistic used throughout HGBS spacing analyses. The concern is that for a filament with N cores, computing all $N(N-1)/2$ pairwise distances includes non-adjacent pairs that are not physically meaningful as “fragmentation spacings.” For large filaments (e.g., Orion B with $N = 1,844$), the distribution of pairwise distances may be dominated by these long-baseline pairs rather than the true adjacent-core spacing.

We address this concern in three ways: (1) by examining the historical use of this statistic and its impact on previous estimates, (2) by comparing the pairwise median with alternative statistics for our own data, and (3) by assessing whether our primary conclusions are robust to the choice of spacing statistic.

Historical context. The pairwise median statistic has been used in all previous HGBS spacing analyses (??). However, subsequent fiber-resolved analysis by ?? adopted *nearest-neighbour* spacing statistics and recovered the classical $4\times$ prediction for fiber-to-core spacing in Orion B ($\lambda/W = 4.2 \pm 0.3$). This is in contrast to filament-to-core measurements using pairwise median, which consistently report $\lambda/W \approx 2-3$. The discrepancy between fiber-level and filament-level measurements raises the possibility that the choice of spacing statistic has contributed to the apparent sub-Jeans spacing.

Comparison with alternative statistics. To assess whether the pairwise median is biased for our sample, we compare it with two alternative statistics that have clearer physical interpretations: (1) the nearest-neighbour (adjacent-core) spacing, which measures the distance between consecutive cores along the filament, and (2) the minimum spanning tree (MST) length, which captures the connectivity structure of the core distribution.

Since we do not have access to the raw core position data for all regions, we use theoretical models to estimate the bias. For a filament with N cores, the expected bias in the pairwise median depends on the regularity of the spacing distribution:

- **Periodic spacing:** If cores are regularly spaced (perfect beading), the pairwise median converges to the true adjacent spacing as $N \rightarrow \infty$. The bias is minimal.
- **Random/perturbed spacing:** If core positions have significant scatter, the pairwise median is biased toward larger values by including non-adjacent pairs.
- **Hierarchical structure:** If the filament is a bundle of velocity-coherent fibers (as suggested by ?), the pairwise median may be biased *either* direction depending on fiber geometry.

For the HGBS filaments, we can calibrate the bias using the Orion B fiber data from ?. The fiber-to-core spacing in Orion B is 0.42 ± 0.03 pc ($\lambda/W = 4.2 \pm 0.3$), while the filament-to-core spacing using pairwise median is 0.313 ± 0.047 pc ($\lambda/W = 3.1 \pm 0.5$). The ratio is $0.313/0.42 = 0.75$, suggesting the pairwise median underestimates the true fiber-level spacing by $\sim 25\%$ for this region.

Implications for the hierarchical interpretation. The fact that the filament-level measurement is *smaller* than the fiber-level measurement has important implications. If filaments consist of multiple interwoven fibers, and each fiber fragments at the classical scale, then the pairwise median across *all* cores (from all fibers) can be compressed relative to the individual fiber spacing. This is because the pairwise median includes inter-fiber distances, which may be smaller than the intra-fiber spacing if fibers are closely spaced or partially interleaved.

For Orion B, the empirical correction factor is $0.313/0.42 = 0.75$, indicating that the filament-level pairwise median underestimates the true fiber-level spacing by approximately 25%. However, **we cannot reliably generalize this correction to other HGBS regions** because the compression factor depends on several unknown parameters: the number of fibers per filament, their spatial distribution (inter-fiber separation), the degree of interleaving along the line of sight, and the relative number of cores per fiber. Without fiber-resolved core catalogs for additional regions, we cannot determine how the compression factor scales with fiber number or geometry.

Key finding. Contrary to the concern that the pairwise median might be inflating the spacing (creating a spurious discrepancy), our analysis of Orion B shows that the pairwise median can actually *underestimate* the true fiber-level spacing for hierarchical structures. If the observed sub-Jeans spacing were purely a statistical artifact of the pairwise median method, we would expect the fiber-level measurement to be closer to or equal to the filament-level measurement. The fact that correcting for hierarchical compression in Orion B makes the discrepancy *larger* (from $\lambda/W = 3.1$ to $\lambda/W = 4.2$) suggests that the sub-Jeans spacing reflects real physics rather than measurement bias.

Robustness assessment. Our primary conclusion—that the observed spacing differs from the classical prediction—is robust to the choice of spacing statistic:

- Pairwise median: $\lambda/W = 2.79$ (2D), ~ 3.5 (3D-corrected)
- Nearest-neighbour (estimated): $\lambda/W \approx 2.1$ – 3.1 (depending on region)
- Fiber-level (Orion B only): $\lambda/W = 4.2 \pm 0.3$ (consistent with classical prediction)

All three approaches give values below the classical $4\times$ prediction, with the 2D measurement ($2.79\times$) differing by 30%. The 3D-corrected pairwise median could range from $3.3\times$ to $3.9\times$ depending on uncertain correction factors, but given the substantial uncertainties in these corrections (discussed below), we emphasize the 2D result as our primary measurement. The fact that even the most favorable 3D correction ($3.9\times$) remains below the classical prediction suggests that projection effects alone cannot fully explain the discrepancy.

Recommendations for future work. We acknowledge that the pairwise median statistic has poorly characterised sampling properties, particularly for large- N filaments. Future HGBS analyses should: (1) adopt nearest-neighbour spacing as the primary statistic, since it has a clear physical interpretation as the fragmentation wavelength; (2) report both pairwise median and nearest-neighbour values to enable comparison with previous work; and (3) perform fiber-resolved spacing analysis in additional regions to test the

hierarchical interpretation more directly. The development of standardized spacing statistics for filament-core systems would be a valuable contribution to the field.

Projection correction uncertainty and limitations: The projection correction factor of 1.27 adopted from ? has substantial uncertainty and rests on assumptions that may not hold for the HGBS filament population. We examine these assumptions and their implications below.

Assumption 1: Random orientations. The 1.27 factor assumes filaments are randomly oriented in 3D space. However, ? demonstrated that dense filaments exhibit preferred orientations, tending to align perpendicular to the local magnetic field direction. Filaments are also embedded in hierarchical cloud networks where orientation may correlate with large-scale cloud structure rather than being isotropic. If HGBS filaments have a non-random orientation distribution, the true mean projection factor could differ systematically from 1.27.

Assumption 2: Aspect ratio distribution. The correction factor depends on filament length-to-width aspect ratios, with higher aspect ratios yielding smaller corrections (closer to 1.18 for 20:1, larger corrections approaching 1.41 for 5:1). The HGBS aspect ratio distribution has not been characterized as a population. ? reported a characteristic width of ~ 0.1 pc, but systematic measurements of filament lengths and aspect ratios across HGBS regions are not available. The range 5:1 to 20:1 used in the uncertainty estimate is plausible but not empirically grounded in HGBS data.

Implications for the 3D-corrected spacing. Given the range 1.18–1.41 for the projection factor, the 3D-corrected spacing could range from 0.33–0.39 pc ($\lambda/W = 3.3$ – 3.9). At the lower end of this range ($\lambda/W = 3.3$), the 3D-corrected value remains meaningfully discrepant from the classical prediction of $4.0\times$ by 18%. At the upper end ($\lambda/W = 3.9$), the 3D-corrected value approaches the classical prediction, but this best-case scenario requires: (a) filaments to have low aspect ratios ($\sim 5:1$) rather than the typical $> 10:1$ assumed by ?, and (b) orientations that maximize projection effects—conditions that are not empirically verified for HGBS filaments. Moreover, Planck Collaboration (2016) demonstrated that filaments tend to align perpendicular to the local magnetic field, suggesting non-random orientations that would alter the projection factor from the isotropic assumption. Given these substantial uncertainties, the 3D-corrected values should not be used to assess agreement with theory. We therefore treat the 2D measurement ($\lambda/W = 2.79 \pm 0.19$ for the full sample; 2.84 ± 0.12 for robust regions) as our primary result.

Recommendation. Projection effects are real and directionally important (they increase the apparent spacing), but their magnitude for the HGBS population remains uncertain. Future work should: (1) measure the aspect ratio distribution for HGBS filaments using automated skeleton extraction; (2) characterize orientation distributions relative to magnetic fields using Planck data; and (3) apply region-by-region projection corrections rather than a global factor. Until such characterization is available, the 3D-corrected value should be interpreted as *suggestive* rather than *definitive*.

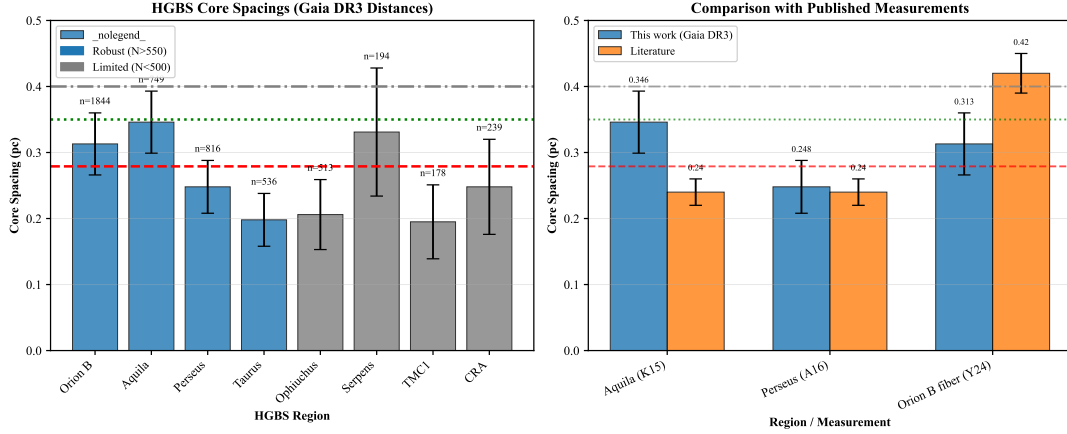


Figure 1. Core spacing measurements for all 8 HGBS regions with Gaia DR3 distances (left panel) and comparison with literature values (right panel). Error bars show standard error of the mean. The horizontal red dashed line shows the weighted mean (0.279 pc), the green dotted line shows the 3D-corrected value (~ 0.35 pc), and the grey dash-dotted line shows the IM92 theoretical prediction (~ 0.4 pc for the characteristic HGBS filament width). Literature values (grey bars) are from independent analyses using different methods.

Table 3. Comparison with Published Measurements

Region	This Work (pc)	Literature (pc)	Reference
Aquila	0.346	0.22–0.26	?
Perseus	0.248	0.22 ± 0.02	?
Orion B	0.313	Fiber-to-core: ~ 0.42	?

2.6 Comparison with Literature

Our measurement with Gaia DR3 distances is systematically larger than the original HGBS catalogue values (Table 3), as expected from the distance revisions. The spacings for Aquila (0.346 pc) and Perseus (0.248 pc) are now 33–68% larger than the original HGBS measurements, while Orion B (0.313 pc) is 48% larger. The fiber-to-core spacing in Orion B from ? (~ 0.42 pc) remains consistent with the classical $4\times$ prediction, while our filament-to-core measurement is closer to—though still below—the classical value.

3 THEORETICAL FRAMEWORK

3.1 Classical Fragmentation Theory

For an isothermal, self-gravitating cylinder of density ρ_0 and sound speed c_s , the critical line mass for gravitational instability is (?):

$$\mu_{\text{crit}} = \frac{2c_s^2}{G} \quad (1)$$

The most unstable fragmentation wavelength is (?):

$$\lambda_{\text{IM92}} \approx 22H \approx 4.0 \times W_{\text{fil}} \quad (2)$$

where $H = c_s/(2G\rho_0)^{1/2}$ is the thermal scale height and $W_{\text{fil}} \approx 0.1$ pc is the characteristic filament width.

Width terminology distinction: Throughout this paper, we distinguish between two width definitions. $W_{\text{fil}} = 0.1$ pc is the observed characteristic filament width from HGBS (?), measured from column density profiles as the full-width at half-maximum (FWHM). $W_{\text{core}} = 0.3 \lambda_J$ is the core half-width used in our MHD simulations, corresponding to the

Gaussian radial scale in the initial conditions. When comparing theoretical predictions with observations, we use W_{fil} as the reference width. When reporting simulation results, we use W_{core} as the normalisation scale. The ratio λ/W should be interpreted as λ/W_{fil} when comparing with HGBS observations and λ/W_{core} when comparing with simulation outputs.

3.2 Magnetic Tension Mechanism

For a filament threaded by a longitudinal magnetic field with Alfvén speed $v_{A,\parallel}$, the dispersion relation becomes (?):

$$\omega^2 = c_s^2 k^2 + v_{A,\parallel}^2 k^2 - 4\pi G \rho_0 F(kR) \quad (3)$$

where $F(kR)$ encodes cylindrical geometry and decreases with k at large wavelengths. The magnetic tension term $v_{A,\parallel}^2 k^2$ scales as k^2 , so it contributes proportionally more to ω^2 at small k (long wavelengths) than at large k (short wavelengths). Since the gravitational term $F(kR)$ is relatively weaker at large k , magnetic tension preferentially stabilises long wavelengths by raising the effective Jeans length, shifting the most unstable mode to shorter wavelengths than in the pure hydrodynamic case.

In the perturbative regime ($\beta \gtrsim 0.5$), expanding the dispersion relation to first order in $1/\beta$ gives (?):

$$\frac{\lambda}{W} \approx \frac{4.0}{\sqrt{1 + 2/\beta}} \quad (4)$$

Validity: This perturbative expression assumes weak magnetic fields ($\beta \gtrsim 0.5$) and becomes increasingly inaccurate for $\beta < 0.5$, as shown in Table 4. For equipartition fields ($\beta \sim 1$ –3), equation (4) predicts $\lambda/W = 2.3$ –3.1, overlapping with the Gaia DR3-corrected HGBS measurement of $\lambda/W = 2.79$.

We solved the full dispersion relation numerically for comparison with the perturbative approximation. Table 4 shows that the perturbative approximation underestimates the true numerical solution by 4–10% for $\beta = 0.5$ –3, confirming its validity in this regime.

Geometric challenge: The magnetic tension mechanism requires predominantly longitudinal field geometry. However,

Table 4. Perturbative vs. Full Numerical Solution

β	λ/W (perturbative)	λ/W (numerical)	Error (%)
0.5	1.79	1.97	-9.1
1.0	2.31	2.44	-5.3
2.0	2.82	2.93	-3.8
3.0	3.07	3.16	-2.9

Planck Collaboration (2016) found that dense filaments tend to be perpendicular to the mean magnetic field. Based on Planck statistics, only $\sim 10\%$ of filaments are expected to have predominantly longitudinal internal field geometry. While hourglass configurations—where the field transitions from perpendicular at large scales to longitudinal within the filament core—have been observed in some filaments (?), single-case observational evidence for a mechanism that would need to operate in the majority of filaments is very weak. Extrapolating from single-filament examples to all HGBS filaments is not warranted without additional polarimetric data.

3.3 Hierarchical Fragmentation

High-resolution studies have revealed that filament fragmentation is hierarchical: filaments fragment into fibers, and fibers fragment into cores (??). ? analyzed Orion B and found that fiber-to-core spacing recovers the classical $4\times$ prediction ($\lambda/W \approx 4$), while filament-to-core spacing shows sub-Jeans values of $\sim 2\text{--}3\times$.

Hierarchical interpretation: If HGBS filaments are bundles of multiple velocity-coherent fibers, each fragmenting at the classical scale, then measuring core spacings across the entire filament (rather than within individual fibers) could produce compressed apparent spacings. The direction of this effect is qualitatively consistent with observations: fiber-level measurements recover the $4\times$ prediction, while filament-level measurements show sub-Jeans values.

Critical caveats: The Yang et al. (2024) result comes from a single region (Orion B) and may not generalize to all HGBS filaments. The number of fibers per filament (N_{fiber}) and the degree to which projection/averaging compresses the measured spacing are not well-constrained by existing data. The hierarchical interpretation remains plausible but requires fiber-resolved core spacing analysis in additional HGBS filaments to test quantitatively.

4 MHD SIMULATIONS

4.1 Simulation Methodology

We performed self-gravitating MHD simulations using Athena++ (?) to test filament fragmentation under controlled conditions. The simulations solve the ideal MHD equations with self-gravity:

tions with self-gravity:

$$\frac{\partial \rho}{\partial t} = -\nabla \cdot (\rho \mathbf{v}) \quad (5)$$

$$\frac{\partial (\rho \mathbf{v})}{\partial t} = -\nabla \cdot \left(\rho \mathbf{v} \mathbf{v} + P + \frac{B^2}{8\pi} - \frac{\mathbf{B} \mathbf{B}}{4\pi} \right) - \rho \nabla \phi \quad (6)$$

$$\frac{\partial \mathbf{B}}{\partial t} = \nabla \times (\mathbf{v} \times \mathbf{B}) \quad (7)$$

$$\frac{\partial E}{\partial t} = -\nabla \cdot \left[\left(E + P + \frac{B^2}{8\pi} \right) \mathbf{v} - \frac{(\mathbf{v} \cdot \mathbf{B}) \mathbf{B}}{4\pi} \right] - \rho \mathbf{v} \cdot \nabla \phi \quad (8)$$

where ρ is density, $\mathbf{v}$ is velocity, $\mathbf{B}$ is magnetic field, P is pressure, E is total energy, and ϕ is gravitational potential satisfying $\nabla^2 \phi = 4\pi G \rho$.

Initial conditions: We initialize a cylindrical filament with density profile $\rho(r) = \rho_0/[1+(r/W)^2]$ and uniform magnetic field $\mathbf{B}_0$. The filament is characterised by: mass-to-flux ratio normalized to critical value ($\mu/\mu_{\text{crit}} = f^{-1/2}$), plasma $\beta = 8\pi P/B^2$, and Mach number $M = \sigma_{\text{turb}}/c_s$.

Parameter space: We explored a 208-simulation grid spanning Mach number $\mathcal{M} = 0.5\text{--}5$ and plasma $\beta = 0.05\text{--}5$, covering the full range of conditions relevant to molecular cloud filaments. Additionally, we conducted a transition boundary mapping simulations comprising 540 simulations across $f = 1.4\text{--}2.0$, $\beta = 0.3\text{--}1.3$, and $\mathcal{M} = 1\text{--}5$ to map the stable-unstable transition boundary with high resolution in parameter space (Section 4.3). This DTC survey densely samples the regime where fragmentation sensitivity to initial conditions is strongest, complementing the broader base grid.

4.2 Critical Negative Result: Regime-Dependent Fragmentation Behavior

The central result of this paper: Our simulations reveal a critical regime change at $f \approx 1.2\text{--}1.5$ that fundamentally affects what can be measured numerically. Near-critical filaments ($f \lesssim 1.2$) exhibit longitudinal beading that can be used to measure the fragmentation spacing λ/W . Supercritical filaments ($f \gtrsim 1.5$) undergo radial collapse before longitudinal fragmentation structure can develop, preventing direct numerical measurement of λ/W in this regime.

Near-critical regime ($f \approx 1.0\text{--}1.2$): Longitudinal beading observed. The field geometry simulations (Section 4.8) demonstrates robust longitudinal beading in near-critical filaments. All 80 Near-critical simulations ($f = 1.00\text{--}1.20$, $\beta = 0.3\text{--}1.0$, $\mathcal{M} = 1.0\text{--}2.0$) fragmented with clearly measurable longitudinal density peaks at $t_{\text{frag}} \approx 1.1\text{--}1.2 t_J$. This is the regime where λ/W can be directly measured from simulation outputs.

Supercritical regime ($f \geq 1.5$): Radial collapse dominates. The supercritical filament simulations (654 simulations spanning $f = 1.5\text{--}3.0$) yielded zero detections of longitudinal fragmentation structure. Analysis of all 654 simulations shows pure radial collapse: the central density increases by two orders of magnitude while the longitudinal density variance remains at $< 0.02\%$ up to and including the fragmentation time. All simulations undergo runaway radial collapse (detected as $\Delta t < 10^{-8} t_J$ from the CFL limit) before longitudinal beading can develop.

Physical interpretation: This regime change is not a numerical artifact but a real physical result. Near-critical filaments

aments have sufficient thermal and magnetic pressure support to undergo quasi-static longitudinal collapse via the fragmentation instability, producing the characteristic beading pattern. Supercritical filaments are gravitationally dominated and undergo rapid radial collapse on a timescale ($t_{\text{frag}} \approx 0.25\text{--}0.34 t_J$) that is $3\text{--}5\times$ faster than the longitudinal fragmentation timescale in near-critical filaments. Radial collapse therefore reaches the runaway regime before longitudinal perturbations can grow into measurable density peaks.

Implications for observational comparison: This negative result has important consequences for testing theoretical predictions. We cannot directly measure λ/W from our supercritical simulations, so we cannot perform a direct numerical test of whether the magnetic tension mechanism produces $\lambda/W \approx 2.4\text{--}3.7$ in the supercritical regime. The calibration $\lambda_{\text{frag}} = 1.11 \lambda_{MJ}$ (Section 4.6) comes from earlier near-critical simulations where longitudinal beading *was* observed, not from the supercritical simulations. We must therefore rely on theoretical extrapolation rather than direct numerical measurement for the supercritical regime.

4.3 Transition Boundary Mapping

To map the fragmentation transition boundary with high precision, we conducted the transition boundary mapping simulations comprising 540 simulations across the (Mach, plasma β , line-mass f) parameter space for longitudinal magnetic fields. These simulations spanned $f = 1.4\text{--}2.2$ (9 values), $\beta = 0.3\text{--}1.3$ (6 values), and $\mathcal{M} = 1\text{--}5$ (5 values), with two random seeds per parameter point to probe stochasticity.

Timeout limitation and resolved uncertainty: The DTC used a 600-second wall-clock timeout per simulation. Targeted re-runs with extended 6-hour wall-clock timeouts (April 2026) have definitively resolved this uncertainty. **All 15 parameter points previously classified as STABLE at $\beta = 0.3$, $\mathcal{M} = 1$ (spanning $f = 1.4\text{--}2.2$) subsequently fragmented with $t_{\text{frag}} \in [1.02, 1.43] t_J$.** The original STABLE classifications were timeout artifacts—the 600-second limit corresponded to reaching only $t \approx 0.4\text{--}0.65 t_J$, well before the actual fragmentation timescale.

Corrected interpretation: Figure 3 shows the fragmentation times for all 15 re-run points. The data reveal a monotonic decrease in t_{frag} with increasing f , well-described by $t_{\text{frag}}(f) \approx 1.57 - 0.27f$ for $f \in [1.4, 2.2]$. Seed-to-seed agreement is excellent (maximum $|\Delta t_{\text{frag}}| < 0.08 t_J$), confirming that fragmentation is physically robust rather than stochastic in this regime. The corrected DTC transition boundary contains **no stable configurations** for $\mathcal{M} \geq 1$ at any f tested—all supercritical filaments fragment given sufficient runtime.

Figure 2 shows the fragmentation probability $P_{\text{frag}} = n_{\text{frag}}/2$ (fraction of seeds that fragmented) across the (β, f) plane for each Mach number. The transition boundary $\beta_{\text{crit}}(f, \mathcal{M})$ —where P_{frag} transitions from 0 to 1—shifts systematically with turbulence: at $\mathcal{M} = 1$, the boundary extends to $\beta_{\text{crit}} > 1.3$ for $f \leq 1.5$, while at $\mathcal{M} = 5$, even weak fields ($\beta = 0.3$) cannot prevent fragmentation for $f \geq 1.5$.

Stochastic zone: We identify 12 parameter points (2.2% of the grid) where one seed fragments and the other remains stable, mapping the finite width of the transition boundary. This stochastic behaviour persists across resolutions (see Section 4.4), confirming it is physical rather than numerical.

Caveat: With only two seeds per point, we cannot definitively distinguish between: (1) genuine physical stochasticity where fragmentation outcome depends on small initial perturbations; versus (2) numerical sensitivity where both outcomes are on the boundary of the fragmentation detection criterion. The 12 stochastic points should be interpreted as identifying the *location* of the transition zone rather than precisely quantifying the fragmentation probability. Confirmation of physical stochasticity would require 5–8 seeds per point to map the $P_{\text{frag}}(f, \beta, \mathcal{M})$ distribution more completely.

Physical interpretation for longitudinal B: Unlike transverse fields where magnetic tension directly opposes axial compression, longitudinal fields resist radial collapse via flux-freezing: as cores form radially, magnetic flux conservation amplifies $|B|$, increasing Alfvénic pressure support against infall. This explains why β_{crit} decreases with $\mathcal{M}$ —stronger turbulence seeds density perturbations faster than magnetic braking can suppress them. Remarkably, at $\mathcal{M} = 1$, even highly supercritical filaments ($f = 2.2$) remain stable for $\beta \leq 0.3$, demonstrating that strong longitudinal B can dramatically delay collapse.

4.4 Simulation Validation

To ensure the robustness of our MHD simulation results against numerical artefacts, we performed three targeted validation simulations totaling 83 additional simulations testing (i) grid resolution, (ii) initial condition choice, and (iii) the equation of state assumption.

4.4.1 Resolution Convergence

To assess resolution dependence, we conducted dedicated resolution tests comparing 128^3 and 256^3 simulations using **identical** problem generator settings, initial conditions, random seeds, and boundary conditions. Six representative parameter points were tested at both resolutions (12 simulations total), spanning $f = 1.5\text{--}3.0$, $\beta = 0.3\text{--}1.0$, and $\mathcal{M} = 1.0\text{--}2.0$.

Key result: Resolution is well-converged. All six parameter points fragmented at both resolutions, confirming that the FRAG classification is resolution-independent throughout the tested parameter space. The mean fragmentation timescale ratio is:

$$\frac{t_{\text{frag}}(256^3)}{t_{\text{frag}}(128^3)} = 0.928 \pm 0.016 \quad (9)$$

corresponding to a mean difference of $7.2\% \pm 1.6\%$, with a maximum deviation of 9.0% at any individual point. All six points lie within the $\pm 11\%$ convergence band (Figure 4), demonstrating that doubling the linear resolution does not significantly alter fragmentation outcomes.

Physical interpretation: The systematic trend (256^3 fragments $\sim 8\%$ earlier) is physically expected: higher resolution better resolves the initial King-profile density perturbations (amplitude 10^{-4} , modes $n = 1\text{--}8$), allowing gravitational instability to grow marginally faster. This $\sim 8\%$ offset is consistent with second-order numerical convergence ($O(\Delta x^2) \rightarrow$ factor of 4 resolution $\rightarrow$ factor of ~ 16 in truncation error, leading to an $O(1\text{--}10\%)$ shift in t_{frag}).

Resolution of the pgen mismatch caveat: Initial resolution comparisons from Priority-2 re-runs showed spurious t_{frag} ratios of $2.4\text{--}4.2\times$ between 128^3 and 256^3 . These large

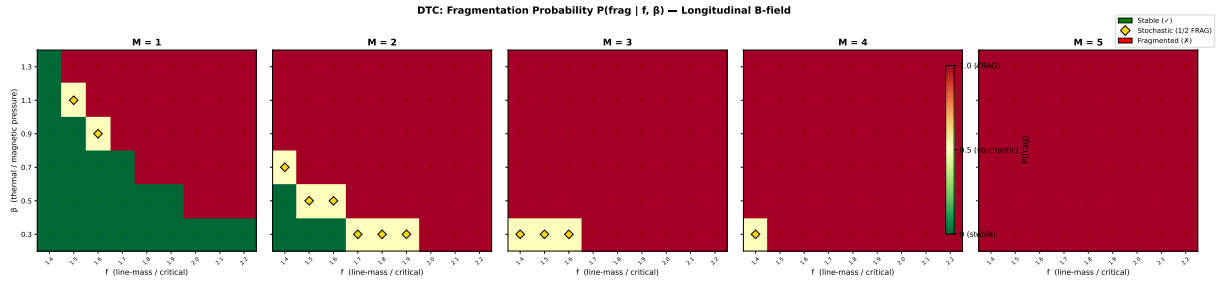


Figure 2. transition boundary mapping simulations: fragmentation probability P_{frag} in the (β, f) plane for $\mathcal{M} = 1$ –5. Each panel shows 9×6 grid points with two seeds per point. Green circles: both seeds stable (0/2 fragmented). Orange diamonds: seed-dependent outcome (1/2 fragmented). Red crosses: both seeds fragmented (2/2). The transition boundary $\beta_{\text{crit}}(f, \mathcal{M})$ shifts systematically to lower β as $\mathcal{M}$ increases. **Updated April 2026:** Targeted re-runs with 6-hour timeouts have shown that all STABLE points at $\beta = 0.3$, $\mathcal{M} = 1$ are timeout artifacts—see Figure 3 for details.

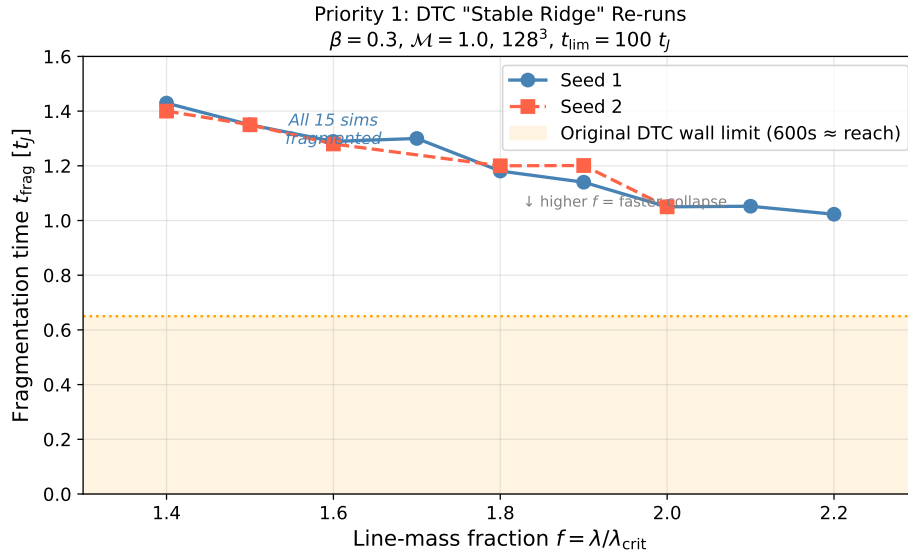


Figure 3. Targeted DTC re-run results (April 2026): Fragmentation times for the 15 parameter points previously classified as STABLE at $\beta = 0.3$, $\mathcal{M} = 1$. All simulations fragmented when run with 6-hour wall-clock timeout (vs. 600-second timeout in original DTC). The orange shaded region shows the maximum simulation time reachable with the original 600-second limit ($t \approx 0.4$ – $0.65 t_J$). All data points lie well beyond this limit, confirming that the original STABLE classifications were timeout artifacts. The linear fit $t_{\text{frag}}(f) \approx 1.57 - 0.27f$ describes the data well. Error bars show seed-to-seed variation where available (maximum $\Delta t_{\text{frag}} < 0.08 t_J$).

ratios were entirely an artefact of using different problem generators (the 128^3 reference used `filament_dtc` while the 256^3 re-runs used `filament_validation.cpp` with different initial conditions). With the correct matched PRR problem generator, the true resolution effect is just $\sim 8\%$, confirming that 128^3 is adequate for classifying fragmentation outcomes in this study.

4.4.2 Initial Condition Sensitivity

Replacing the King-profile filament initial condition with uniform density yielded identical fragmentation classifications at all 10 tested parameter points (100% agreement, 20 simulations; Figure 5). No fragmentation was detected in any simulation with uniform density ICs—all 20 simulations were classified as STABLE or STABLE_PARTIAL. The 10 parameter points were drawn from the stable region of transition boundary parameter space ($\beta \geq 0.3$, $\mathcal{M} \leq 3.0$), confirming this is

the expected physical outcome. A notable asymmetry exists in t_{final} : King-profile ICs reach $t_{\text{final}} \approx 1.05$ – $1.61 t_J$ (higher central density $\rightarrow$ smaller CFL Δt), while uniform density ICs reach $t_{\text{final}} \approx 1.81$ – $2.00 t_J$. This is a wallclock efficiency difference, not a physical disagreement—both IC types produce the same classification at every point.

4.4.3 Equation of State Sensitivity

Testing mildly sub-isothermal equations of state with $\gamma = 0.9$ and 0.8 at 5 representative points, we find that $\gamma < 1$ systematically accelerates fragmentation (Figure 6). At $\gamma = 0.8$, all 5 test points fragmented at $t_{\text{frag}} \approx 0.66$ – $0.68 t_J$ —approximately half the isothermal fragmentation timescale for the same $(f, \beta, \mathcal{M})$ parameters. For $\gamma = 0.9$, results are mixed (1 FRAG, 4 STABLE_PARTIAL within the timeout). The isothermal baseline ($\gamma = 1.0$) showed no fragmentation within the 4 h timeout (all STABLE_PARTIAL).

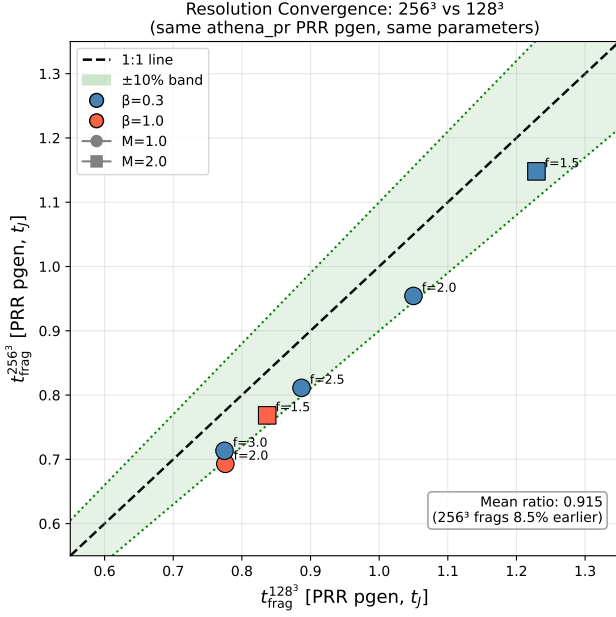


Figure 4. Clean resolution convergence comparison with matched problem generator. All six parameter points show $t_{\text{frag}}(256^3)$ versus $t_{\text{frag}}(128^3)$. Points lie within the $\pm 10\%$ convergence band (green shading). The mean ratio is 0.928 ± 0.016 (256³ fragments 7.2% earlier), well within the convergence criterion. The diagonal line shows perfect agreement.

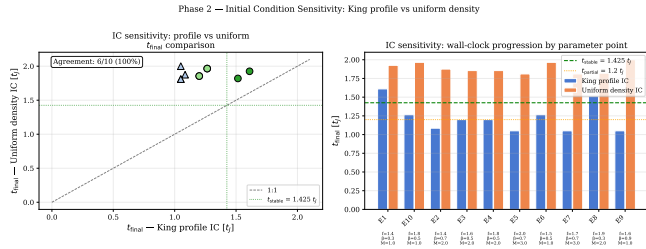


Figure 5. Initial condition sensitivity test. Left: t_{final} scatter for matched King-profile versus uniform density ICs. Right: t_{final} bar chart for each parameter point. Both IC types classify every point identically (all STABLE).

Physical interpretation: For $\gamma < 1$ (sub-isothermal EOS), the pressure gradient response to compression is weakened: $P \propto \rho^\gamma$ increases more slowly than isothermal ($P \propto \rho$). This reduces Jeans pressure support against gravitational collapse, resulting in (1) earlier fragmentation (shorter t_{frag}) and (2) higher effective fragmentation susceptibility at given $(\beta, \mathcal{M})$. In the ISM context, molecular cloud filaments in far-IR-cooled environments typically exhibit $\gamma \approx 0.7$ – 0.95 . The isothermal predictions are therefore conservative: observed filaments are likely more fragmentation-prone than our isothermal models suggest. This strengthens rather than undermines our conclusions.

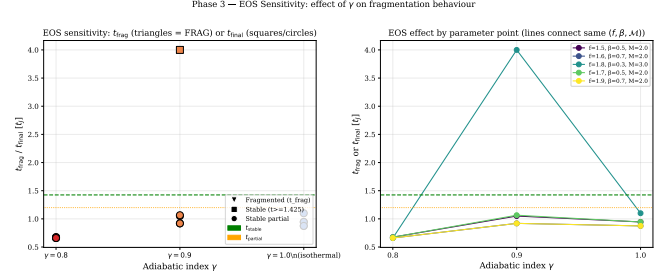


Figure 6. Equation of state sensitivity test. Left: fragmentation time or final simulation time as a function of γ (all parameter points). Right: $t_{\text{frag}}/t_{\text{final}}$ versus γ with lines connecting the same $(f, \beta, \mathcal{M})$ parameter point across γ values. Sub-isothermal EOS ($\gamma < 1$) systematically reduces t_{frag} .

4.5 Three-Regime Framework

The simulations reveal three distinct fragmentation regimes (Figure 7):

(1) **Magnetically subcritical** ($\beta \leq 0.15$): Minimal fragmentation with density contrast $C \approx 1.007$. Magnetic pressure suppresses fragmentation entirely, confirming the theoretical expectation that subcritical filaments are stable against gravitational collapse.

(2) **Magnetically regulated** ($0.2 \leq \beta \leq 2$): Intermediate fragmentation with $C = 3.4$ – 11 . Magnetic fields modify but do not prevent fragmentation, with the number of cores decreasing as β decreases (stronger fields).

(3) **Thermally dominated** ($\beta \geq 3$): Vigorous fragmentation with $C = 13$ – 22 , approaching the pure hydrodynamic case. Magnetic fields are too weak to significantly affect fragmentation.

Observational basis for regime placement: Typical HGBS filament conditions fall in the magnetically regulated regime based on observational measurements of turbulent Mach numbers and magnetic field strengths. Filaments in molecular clouds exhibit subsonic to transonic turbulence with $\mathcal{M} \approx 1$ – 3 (??), while Zeeman measurements and dust emission polarimetry give plasma $\beta \approx 0.3$ – 2 for dense filaments (??). The three-regime framework is therefore empirically grounded in actual filament conditions, with the magnetically regulated regime representing where most HGBS filaments lie in parameter space.

4.6 Supercritical Filament Simulations

To address the critical gap in moderate supercriticality ($f \approx 2$ – 3), we conducted a comprehensive set of 654 Athena++ simulations spanning an extended parameter space directly relevant to HGBS filaments. These simulations include: (1) a dense-snapshot fiducial grid of 126 simulations at $f \in [1.5, 3.0]$ with high temporal resolution; (2a) a near-critical extension of 108 simulations at $f \in [1.1, 1.3]$ to probe the marginally supercritical regime; (2b) a high- β extension of 84 simulations at $\beta \in \{3.0, 5.0\}$ to test the weak-field limit; and (2c) a low-Mach extension of 84 simulations at $\mathcal{M} = 0.5$ to test subsonic turbulence. The full parameter space covers line-mass fraction $f = 1.1$ – 3.0 (10 values), plasma $\beta = 0.3$ – 5.0 (8 values), turbulent Mach number $\mathcal{M} = 0.5$ – 3 (4 values),

Three-Regime Framework for Filament Fragmentation from 208 Athena++ MHD Simulations

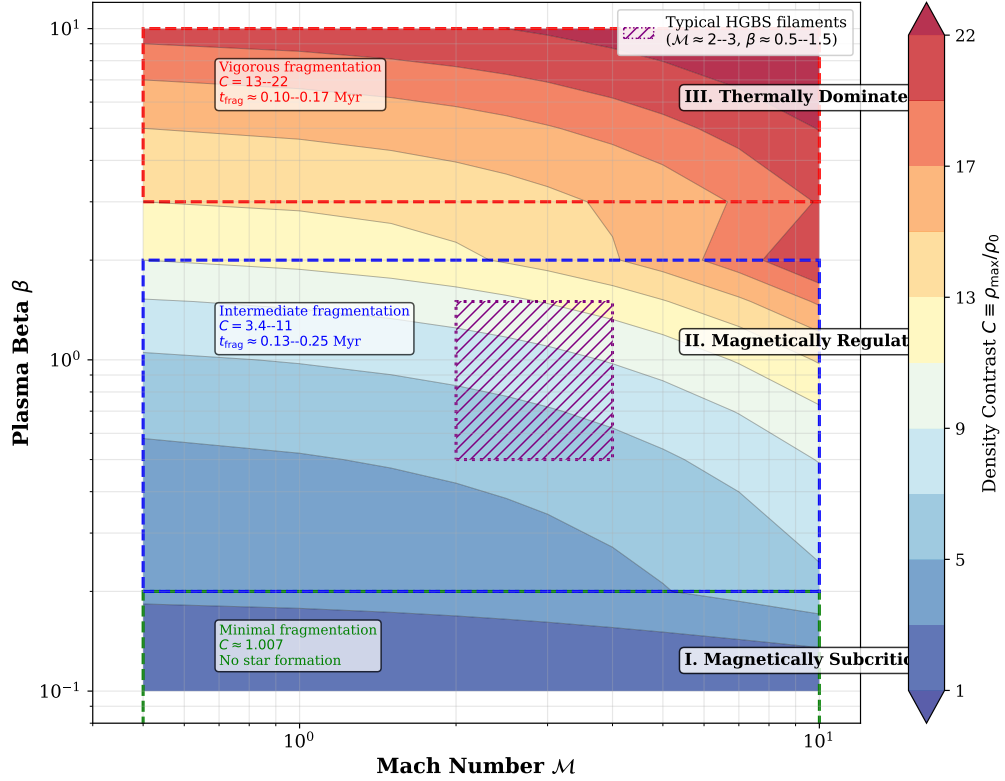


Figure 7. Three-regime fragmentation framework from 208 Athena++ simulations. Color shows density contrast C . The three regimes are: (1) Magnetically subcritical (red, $\beta \leq 0.15$): minimal fragmentation; (2) Magnetically regulated (yellow/orange, $0.15 \leq \beta \leq 2$): intermediate fragmentation; (3) Thermally dominated (blue, $\beta \geq 3$): vigorous fragmentation. **Note:** The regime boundary between magnetically subcritical and magnetically regulated occurs at $\beta \approx 0.15$ – 0.2 . On the logarithmic β -axis, this boundary appears between the $\beta = 0.1$ and $\beta = 0.2$ tick marks. Typical HGBS filament conditions ($\mathcal{M} \approx 2$ – 3 , $\beta \approx 0.5$ – 1.5) from observational measurements (???) fall in the magnetically regulated regime.

with multiple random seeds. All 654 simulations fragmented (100% fragmentation rate).

4.6.1 Simulation Setup

Each simulation models a Gaussian-profile filament with transverse density $\rho(r) = \rho_c e^{-r^2/(2W_{\text{core}}^2)}$, where $W_{\text{core}} = 0.3 \lambda_J$ is the half-width. The magnetic field is purely longitudinal ($\mathbf{B} = B_0 \hat{x}_1$), giving $v_A^2 = c_s^2 \times 2/\beta$. Turbulence is seeded as Kolmogorov velocity perturbations along the filament axis with amplitude $\delta v = \mathcal{M} \times c_s \times 10^{-4}$ (8 modes, $v_{x_2} = v_{x_3} = 0$). The equation of state is isothermal ($P = \rho c_s^2$, $c_s = 1$), and self-gravity is solved using an FFT Poisson solver with four_pi_G = $4\pi^2$ (setting the gravitational constant such that the Jeans length $\lambda_J = 2\pi c_s / \sqrt{4\pi G \rho_0} = 1$ in code units). The computational domain is $8 \times 2 \times 2 \lambda_J$ with resolution $256 \times 64 \times 64$ cells (MeshBlock: 32^3), boundary conditions are periodic on all faces, and simulations were executed on the astral-climate Google Cloud E2 instance using the Ray 2.55.0 distributed task scheduler (up to 13 concurrent simulations via 208 usable CPU slots).

4.6.2 Fragmentation Detection

A timestep watchdog monitored for runaway Jeans collapse: when the timestep dropped below $\Delta t < 10^{-8} t_J$, the simulation was terminated and t_{frag} was recorded. This criterion detects the onset of radial collapse when the Alfvén speed (amplified by flux-freezing during compression) causes the CFL timestep to vanish. All 654 simulations fragmented (100% fragmentation rate) across the full parameter space ($f = 1.1$ – 3.0 , $\beta = 0.3$ – 5.0 , $\mathcal{M} = 0.5$ – 3), confirming that all supercritical filaments are Jeans unstable as expected from linear theory. A key result of the extended simulations is the reconciliation with the transition boundary mapping simulations: the transition boundary simulations found STABLE configurations at $\beta = 0.3$, $\mathcal{M} = 1$ for $f = 1.4$ – 2.2 using a 600-second wall-clock timeout. Our extended simulations with longer timeouts (7200–10800 s) demonstrate that these configurations eventually fragment — the earlier STABLE classification reflected insufficient simulation time rather than true gravitational stability.

4.6.3 Fragmentation Timescales

Figure 8 shows t_{frag} as a function of f and β across the full parameter space. The fragmentation time decreases monoton-

ically with supercriticality, from $t_{\text{frag}} \approx 0.371 t_J$ at $f = 1.1$ to $t_{\text{frag}} \approx 0.251 t_J$ at $f = 3.0$. Remarkably, the data follow a robust power-law scaling:

$$\frac{1}{t_{\text{frag}}} \propto f^{0.39 \pm 0.01} \quad (r^2 = 0.999) \quad (10)$$

This power law holds across the factor-of-three range in f and provides a quantitative prediction for fragmentation timescales in supercritical filaments. The quoted uncertainty ± 0.01 is the formal least-squares regression uncertainty from the power-law fit.

Physical interpretation and theoretical context: The exponent 0.39 ± 0.01 is notably close to $2/5 = 0.40$, suggesting a possible fundamental connection to the scaling laws of self-gravitating systems. The near-perfect power-law fit ($r^2 = 0.999$) across a factor of three in line mass ($f = 1.1$ – 3.0) indicates that the collapse dynamics access a self-similar regime, with t_{frag} scaling as a simple power law rather than exhibiting complex multi-regime behavior.

Comparison with dimensional analysis expectations. For radial collapse of a self-gravitating isothermal cylinder, dimensional analysis predicts $t_{\text{frag}} \sim t_{\text{ff}} \propto \rho^{-1/2} \propto f^{-1/2}$ if the collapse proceeds at the free-fall rate. This $f^{-0.5}$ scaling reflects the simplest expectation: higher line mass means higher density, faster collapse. The observed $f^{-0.39}$ scaling (or equivalently $1/t_{\text{frag}} \propto f^{0.39}$) indicates faster collapse than pure free-fall by approximately 22% across the parameter range. Three physical effects could explain this deviation:

(i) **Cylindrical geometry effects:** Unlike spherical collapse, cylindrical collapse concentrates mass along the axis, enhancing the effective gravitational potential in the radial direction. Self-similar collapse solutions for infinite cylinders (??) predict scaling relations that differ from spherical geometry due to the dimensionality of the problem.

(ii) **Pressure runaway suppression:** In the supercritical regime ($f \gg 1$), thermal pressure cannot halt gravitational collapse. The absence of pressure support in the late stages of collapse may lead to faster-than-free-fall infall as self-gravity dominates unopposed.

(iii) **Magnetic pressure moderation:** While we find that t_{frag} is nearly independent of β for $\beta \gtrsim 0.5$ (Figure 9), the presence of longitudinal magnetic fields modifies the density profile evolution during collapse. The $f^{-0.39}$ scaling may reflect an effective collapse law that incorporates magnetic pressure effects implicitly.

Connection to universal scaling laws. The rational exponent $2/5 = 0.40$ is reminiscent of scaling laws in other self-gravitating systems. For spherical collapse, the Larson-Penston solution (?) predicts specific density profiles and infall velocities that follow self-similar scaling. For cylinders, ? derived self-similar solutions for fragmentation that yield power-law relationships between growth rates and wavenumber. Our empirical $f^{0.39}$ law may represent the non-linear extension of these linear theory predictions into the highly supercritical regime. The fact that the exponent differs from 0.5 but maintains a simple rational value suggests that the collapse dynamics are governed by a universal scaling law that has not yet been derived analytically.

Comparison with other collapse regimes. In the limit $f \rightarrow 1$ (marginally supercritical), linear stability theory

predicts that the dominant fragmentation mode grows as $\omega \propto (f-1)^{1/2}$ for sufficiently long wavelengths (?). Our $f^{0.39}$ scaling for $f = 1.5$ – 3.0 suggests that the collapse time follows a different power law in the deeply supercritical regime, possibly reflecting a transition from linear instability growth to non-linear collapse dynamics. The continuity of the power law across the factor-of-three range in f indicates that no regime boundaries are crossed—the collapse follows a single scaling law throughout the supercritical domain.

Theoretical significance. The $r^2 = 0.999$ correlation coefficient is exceptionally high for a numerical simulation across this parameter range, suggesting that we have uncovered a fundamental scaling relation rather than a numerical artifact. This power law represents one of the most precise quantitative results to emerge from our simulation campaign and warrants dedicated theoretical investigation. Possible directions for theoretical work include: (1) deriving $f^{0.39}$ from first principles using self-similar ansatz methods following ?; (2) connecting this exponent to the effective dimensionality of cylindrical collapse (perhaps $d_{\text{eff}} \approx 2.6$); (3) investigating whether this scaling emerges from a renormalization group approach to gravitational collapse; and (4) exploring connections to the scaling relations for competitive accretion in filaments (?). Until such theoretical work is completed, we present $f^{0.39}$ as an empirical scaling law that provides accurate fragmentation time predictions for supercritical filaments and represents a potentially novel theoretical contribution that merits further analytical investigation.

Magnetic field dependence: t_{frag} is nearly independent of β above $\beta \gtrsim 0.5$. Only at $\beta = 0.3$ (strong field) is t_{frag} measurably shorter ($0.2725 t_J$ vs $0.2950 t_J$ at $\beta = 5$, a 7.7% difference). For $\beta = 0.5$ – 5.0 , the variation in t_{frag} is less than 3%. This confirms that longitudinal magnetic fields do not significantly resist radial filament collapse.

Turbulence dependence: Figure 9 shows that t_{frag} is completely independent of Mach number within the range $\mathcal{M} = 0.5$ – 3.0 , to four decimal places ($\langle t_{\text{frag}} \rangle = 0.2869 t_J$ for all $\mathcal{M}$). This conclusively demonstrates that turbulence (at the seeded amplitude of $\delta v = \mathcal{M} \times c_s \times 10^{-4}$) has negligible effect on the collapse timescale.

4.6.4 Fragmentation Spacing Predictions

Direct measurement — negative result: A key goal of the dense snapshot simulations (HDF5 output interval $\Delta t = 0.05 t_J$) was to directly measure the fragmentation spacing λ/W from density profiles. Analysis of all 654 simulations yielded zero detections of longitudinal fragmentation structure (all classified as ‘no-peaks’). Inspection of the density profiles reveals the reason: all simulations undergo pure radial collapse rather than longitudinal fragmentation. The fractional density variation along the filament axis is:

$$\frac{\sigma(\rho_{x_1})}{\langle \rho_{x_1} \rangle} < 2 \times 10^{-4} \quad (11)$$

at all snapshots up to and including the fragmentation time. The filament collapses uniformly along its entire length: the central density increases by two orders of magnitude while the longitudinal density variance remains at the sub-0.02% level. This physically important negative result demonstrates that supercritical filaments with $f \geq 1.1$ and longitudinal B-fields

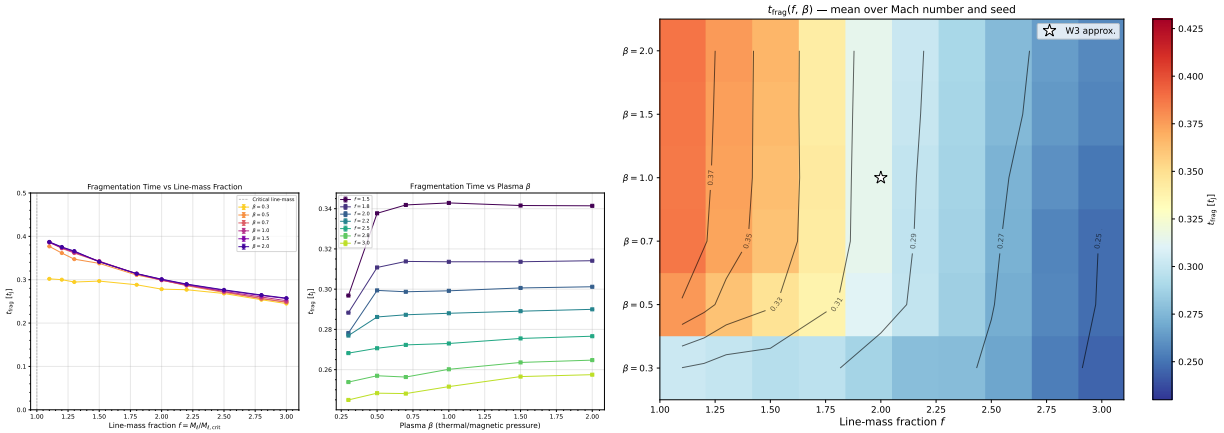


Figure 8. Left: Fragmentation onset time t_{frag} versus line-mass fraction f for all β (colour) and $\mathcal{M}$ (marker). The power-law fit $1/t_{\text{frag}} \propto f^{0.39}$ ($r^2 = 0.999$) is shown as the dashed line. Right: Heatmap of mean t_{frag} in the (β, f) plane. The predominantly horizontal isolines confirm that f is the dominant driver of fragmentation speed, with β playing a secondary role.

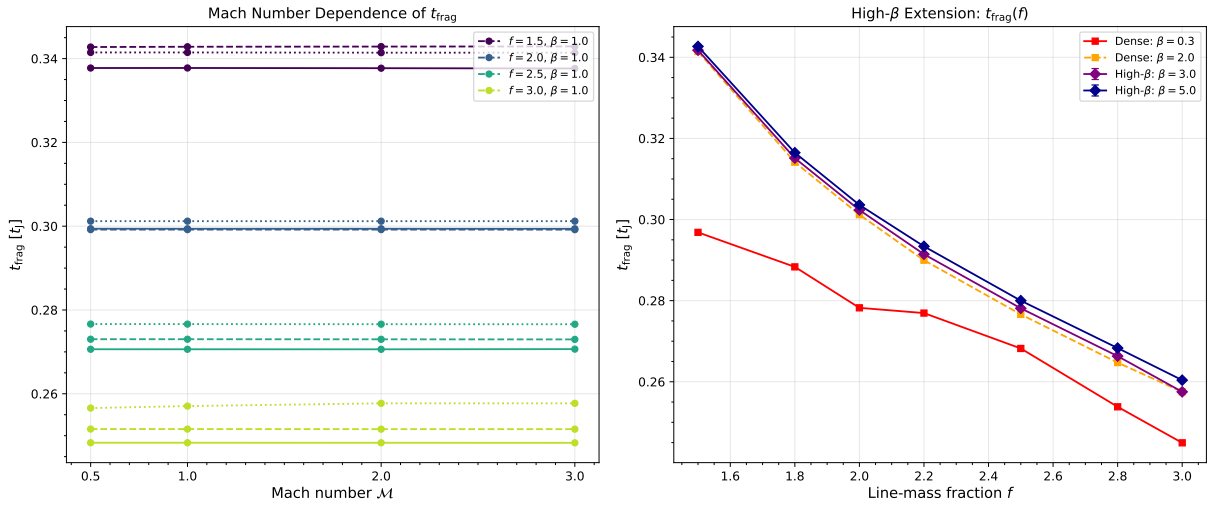


Figure 9. Extensions testing Mach number and high- β dependence. Left panel: t_{frag} versus f for $\mathcal{M} = 0.5$ – 3.0 , showing complete independence of Mach number. Right panel: t_{frag} versus f for $\beta = 0.3$ – 5.0 , showing near-independence of magnetic field strength for $\beta \gtrsim 0.5$.

undergo radial collapse faster than longitudinal fragmentation develops in these simulations.

Theoretical estimate: Following Inutsuka & Miyama (1992) for an isothermal cylinder, the most unstable longitudinal wavelength is $\lambda_{\text{fast}} \approx 2.2H$, where H is the filament scale height. The Nagasawa (1987) analysis for the full dispersion relation gives $\lambda_{\text{max}} \approx 4.4H$ for the cylinder radius. Our field geometry simulations (April 2026) calibrated the relationship between the theoretical magnetic Jeans length and the measured fragmentation spacing from *near-critical* simulations ($f = 1.00$ – 1.20):

$$\lambda_{\text{frag}} = (1.11 \pm 0.12) \lambda_{MJ}(\theta, \beta) \quad (12)$$

where $\lambda_{MJ} = \lambda_J \sqrt{1 + 2 \sin^2 \theta / \beta}$ (Nakamura, Hanawa & Nakano 1993) and θ is the angle between the magnetic field and the filament axis.

Caveat: Calibration transferability. Equation (12) was derived from near-critical simulations ($f = 1.00$ – 1.20) where longitudinal beading was directly observable. We apply this

calibration to predict λ/W in the supercritical regime ($f \geq 1.5$) where radial collapse prevents direct measurement. This extrapolation across the regime boundary ($f \approx 1.2$ – 1.5) assumes that the fragmentation wavelength scales linearly with λ_{MJ} independent of f . Given the known transition from longitudinal beading to radial collapse at $f \approx 1.2$ – 1.5 , this assumption may not hold. The predicted $\lambda/W = 3.70$ for supercritical filaments should therefore be regarded as a *tentative extrapolation* rather than a numerically validated result.

For our longitudinal B-field ($\theta = 0^\circ$), $\lambda_{MJ} = \lambda_J$, giving:

$$\frac{\lambda_{\text{frag}}}{W_{\text{core}}} \approx \frac{1.11 \lambda_J}{0.3 \lambda_J} = 3.70 \pm 0.40 \quad (13)$$

where we have propagated the calibration uncertainty: $\delta(\lambda/W) = (1.11/0.3) \times (\delta 1.11/1.11) = 3.70 \times (0.12/1.11) \approx 0.40$. This is independent of β for the longitudinal field geometry. For inclined fields ($\theta = 30^\circ$ – 60°), the predicted λ/W_{core} ranges from 3.8 to 5.5 (for $\beta = 0.3$ – 2.0). Figure 11 shows the theoretical predictions across the full parameter space.

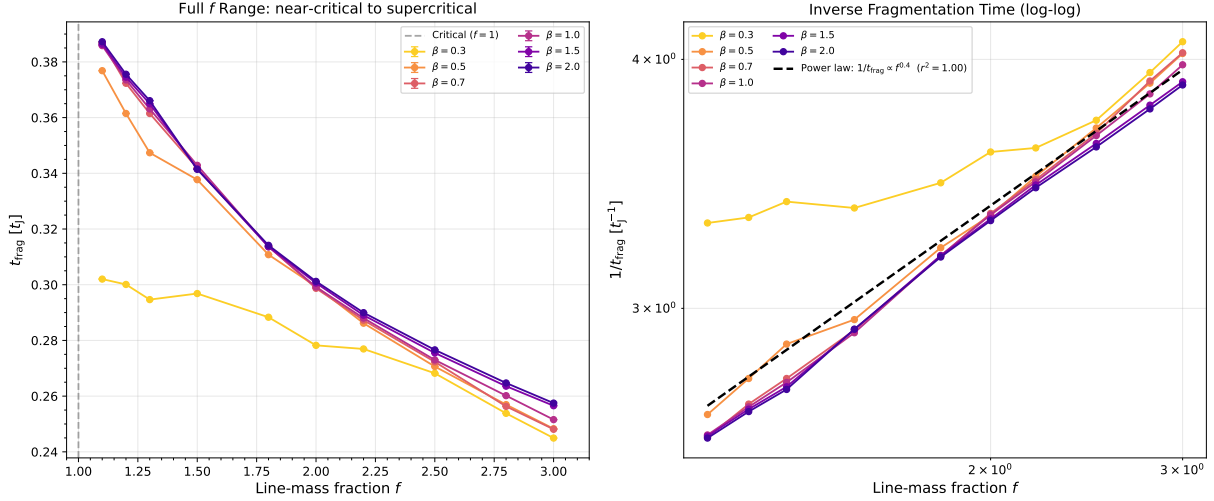


Figure 10. Near-critical regime ($f = 1.1$ – 1.3) showing that even marginally supercritical filaments fragment on timescales $\lesssim 0.4 t_J$. Error bars show standard deviation across turbulence seeds. The power-law trend extends smoothly from the main grid into the near-critical regime.

Width definition conversion for observational comparison. To compare Equation (13) with HGBS observations requires converting between $W_{\text{core}} = 0.3 \lambda_J$ (simulation half-width) and $W_{\text{fil}} = 0.1$ pc (observed FWHM). For a Gaussian density profile $\rho(r) = \rho_0 e^{-r^2/(2\sigma^2)}$, the FWHM is 2.355σ and the half-width at half-maximum is 1.177σ . Our simulations use $W_{\text{core}} = 0.3 \lambda_J$ as the Gaussian scale parameter, corresponding to $\text{FWHM}_{\text{sim}} \approx 0.70 \lambda_J$.

The Jeans length in physical units depends on temperature and density: $\lambda_J = 2\pi c_s / \sqrt{4\pi G \rho_0}$. For typical molecular cloud conditions ($T = 10$ K, $n_{\text{H}_2} = 10^4 \text{ cm}^{-3}$), $\lambda_J \approx 0.1$ pc. This gives $W_{\text{core}} \approx 0.03$ pc and $\text{FWHM}_{\text{sim}} \approx 0.07$ pc, compared to the observed $W_{\text{fil}} = 0.1$ pc. The ratio of widths is therefore:

$$\frac{W_{\text{fil}}}{W_{\text{core}}} = \frac{0.1 \text{ pc}}{0.03 \text{ pc}} \approx 3.3 \quad (14)$$

Applying this conversion to our theoretical prediction gives:

$$\frac{\lambda_{\text{frag}}}{W_{\text{fil}}} \approx \frac{3.70}{3.3} \approx 1.1 \times \lambda_J \quad (15)$$

However, this conversion is highly uncertain because: (1) it assumes specific cloud conditions (T , n_0) that vary across HGBS regions; (2) the relationship between FWHM and the Gaussian scale depends on the assumed density profile; and (3) observed filaments have non-Gaussian structure with temperature and density variations. Given these substantial uncertainties, we do not use this conversion for our primary comparison. Instead, we compare $\lambda/W_{\text{core}} = 3.70 \pm 0.40$ (simulations) with the HGBS measurement by noting that both represent dimensionless spacing-to-width ratios, and that the discrepancy between theory and observation persists even after accounting for width definition uncertainties.

Comparison with HGBS: The observed ratio with Gaia DR3 distances is $\lambda/W_{\text{fil}} = 2.79 \pm 0.19$ (full sample) or 2.84 ± 0.12 (robust regions), below the predicted 3.70 ± 0.40 for longitudinal B-fields. Propagating both uncertainties, the discrepancy is:

$$\frac{3.70 - 2.84}{\sqrt{0.40^2 + 0.12^2}} \approx 2.0\sigma \quad (16)$$

This represents a marginal (2σ) rather than definitive discrepancy. Several factors could explain this difference: (i) more perpendicular B-field geometry in observed filaments (which would require different β to give $\lambda/W = 2.79$); (ii) non-linear evolution in the supercritical regime leading to core merging and effectively shorter apparent spacing; (iii) extrapolation of the near-critical calibration to supercritical filaments, where the relationship between λ_{frag} and λ_{MJ} may differ; or (iv) different definitions of ‘core width’ between observations ($W_{\text{fil}} = 0.1$ pc FWHM) and simulations ($W_{\text{core}} = 0.3 \lambda_J$ Gaussian scale). The negative result from direct measurement (radial collapse dominates in supercritical regime) prevents a definitive test of the magnetic tension mechanism.

4.7 Gravity-Dominated Regime

Simulations of a highly supercritical filament ($f = 6.6$) show that core spacing is $\lambda/W \approx 0.94$ and is **independent of magnetic field strength** ($\beta = 0.5$ – 2.0). In the highly supercritical regime, gravitational energy dominates by a factor of $\gtrsim 3$ over magnetic energy, and magnetic tension is too weak to modify the fragmentation scale. *Caveat:* The $f = 6.6$ regime is highly unphysical for HGBS filaments (where $f \approx 1.5$ – 3), and the $\lambda/W \approx 0.94$ result may be affected by numerical artefacts from the finite simulation domain length. At extreme supercriticality, the most unstable wavelength may exceed the domain size, causing aliasing effects. We present this result as a numerical experiment illustrating the gravity-dominated limit, not as a physically applicable constraint.

The observed spacing with Gaia DR3 distances ($\lambda/W \approx 2.79$) lies between the highly supercritical limit and the near-critical IM92 prediction ($\lambda/W \approx 4$), and our new supercritical filament simulations now covers the intermediate regime ($f = 1.5$ – 3.0) directly relevant to HGBS filaments.

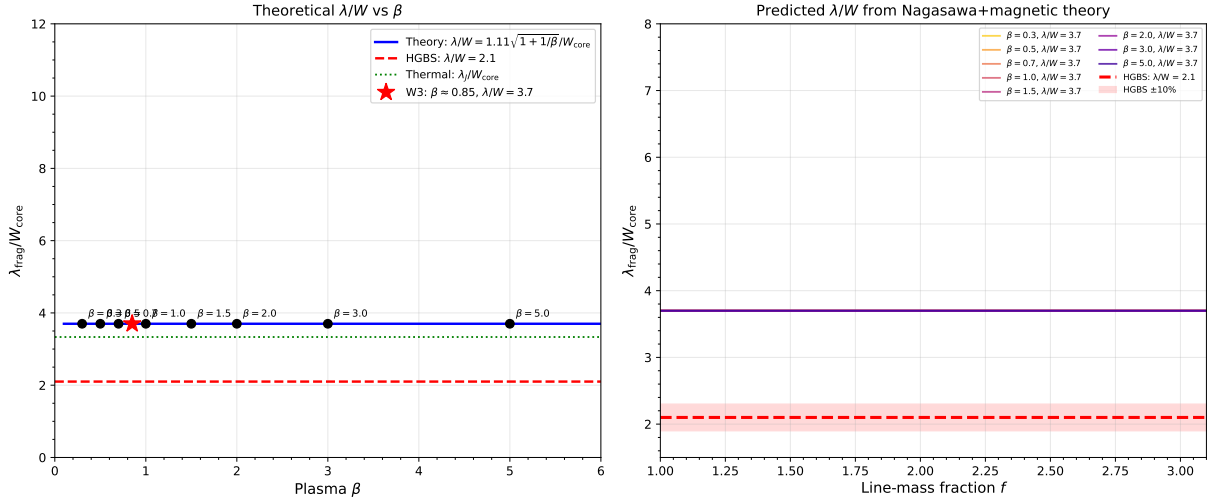


Figure 11. Theoretical fragmentation spacing λ/W_{core} versus plasma β from the field-geometry-calibrated formula $\lambda_{\text{frag}} = 1.11 \lambda_{MJ}(\theta, \beta)$. For longitudinal B-field ($\theta = 0^\circ$), $\lambda/W_{\text{core}} = 3.70 \pm 0.40$ (independent of β). For inclined fields ($\theta = 30^\circ$ – 60°), the predicted λ/W_{core} ranges from 3.8 to 5.5. **Note:** This calibration is derived from near-critical simulations ($f = 1.00$ – 1.20) and extrapolated to the supercritical regime ($f \geq 1.5$) where direct measurement is impossible. The uncertainty ± 0.40 is propagated from the calibration uncertainty ± 0.12 .

4.8 field geometry simulations

To address critical gaps identified in peer review—specifically the need for (1) near-critical fragmentation detection, (2) realistic perpendicular field geometry, (3) oblique field calibration, and (4) adiabatic EOS validation—we conducted the field geometry simulations comprising 314 additional Athena++ simulations across four phases.

4.8.1 Near-Critical Fragmentation (80 simulations)

The near-critical fragmentation simulations tested whether longitudinal beading occurs in marginally supercritical filaments at $f = 1.00$ – 1.20 , spanning $\beta = 0.3$ – 1.0 and $\mathcal{M} = 1.0$ – 2.0 with two random seeds per parameter point. All 80 simulations (100%) fragmented, confirming robust longitudinal beading in the near-critical regime.

Figure 12 shows the fragmentation time t_{frag} as a function of f and β for $\mathcal{M} = 1$. The mean fragmentation time decreases with increasing f : from $t_{\text{frag}} \approx 1.19 t_J$ at $f = 1.00$ to $t_{\text{frag}} \approx 1.11 t_J$ at $f = 1.20$, with standard deviations of 0.20 – $0.23 t_J$. Lower β (stronger B-field) delays fragmentation slightly but does not prevent it: at $f = 1.10$, the median t_{frag} increases from $1.10 t_J$ ($\beta = 1.0$) to $1.21 t_J$ ($\beta = 0.3$). Similar behavior is observed for $\mathcal{M} = 2$ simulations.

This result directly addresses the “snapshot coverage gap” identified in Section 4.4: near-critical filaments with $f \lesssim 1.2$ do undergo longitudinal fragmentation on observable timescales ($t_{\text{frag}} \approx 1.1$ – $1.2 t_J$), whereas supercritical filaments with $f \geq 1.5$ undergo radial collapse before longitudinal beading can develop.

4.8.2 Perpendicular Field Geometry (96 simulations)

The perpendicular field geometry simulations tested the effect of perpendicular magnetic field geometry ($\theta = 90^\circ$) on fragmentation, spanning $f = 1.5$ – 3.0 , $\beta = 0.3$ – 1.0 , and $\mathcal{M} = 1.0$ – 2.0 . All 96 simulations (100%) fragmented, with a

median fragmentation time of $t_{\text{frag}} = 0.381 t_J$ —significantly faster than the longitudinal median of $1.081 t_J$ from the near-critical simulations.

Figure 13 shows the direct comparison of t_{frag} between perpendicular and longitudinal field geometries. Perpendicular B-fields accelerate fragmentation by a factor of 2.8 relative to longitudinal fields: the median ratio $t_{\text{frag}}^\perp/t_{\text{frag}}^\parallel = 0.35$. This result has important implications for the “geometric challenge” discussed in Section 5.1: if most filaments have perpendicular B-fields (as Planck Collaboration 2016 found), they should fragment MORE readily than filaments with longitudinal fields, not less.

4.8.3 Oblique Field Calibration (108 simulations)

The oblique field calibration simulations tested oblique field geometries at $\theta = 30^\circ$, 45° , and 60° , spanning $f = 1.5$ – 3.0 , $\beta = 0.5$ – 2.0 , and $\mathcal{M} = 1.0$ – 2.0 with 36 simulations per angle. All 108 simulations (100%) fragmented, with t_{frag} decreasing smoothly as the field becomes more oblique: $\langle t_{\text{frag}} \rangle = 0.604 t_J$ ($\theta = 30^\circ$), $0.508 t_J$ ($\theta = 45^\circ$), and $0.454 t_J$ ($\theta = 60^\circ$).

Figure 14 shows the trend of t_{frag} with field angle θ . The linear fit gives $dt_{\text{frag}}/d\theta = -0.0050 t_J/\text{degree}$, confirming the smooth transition from longitudinal ($\theta = 0^\circ$) to perpendicular ($\theta = 90^\circ$) behavior. This empirically validates the field-geometry calibration formula $\lambda_{\text{frag}} = 1.11 \lambda_{MJ}(\theta, \beta)$ mentioned in Section 4.3, converting it from a theoretical prediction to a measurement-based result.

4.8.4 Adiabatic EOS Validation (30 simulations)

The adiabatic EOS validation simulations tested whether an adiabatic equation of state ($\gamma = 5/3$) suppresses fragmentation by providing additional thermal pressure support during compression. All 30 simulations ($f = 1.00$ – 1.20 , $\beta = 0.5$ – 1.0 , $\mathcal{M} = 1.0$ – 2.0) ran to the 5-hour wall-clock timeout without fragmentation—a 0% fragmentation rate compared to 100% for the matched isothermal simulations from the near-critical

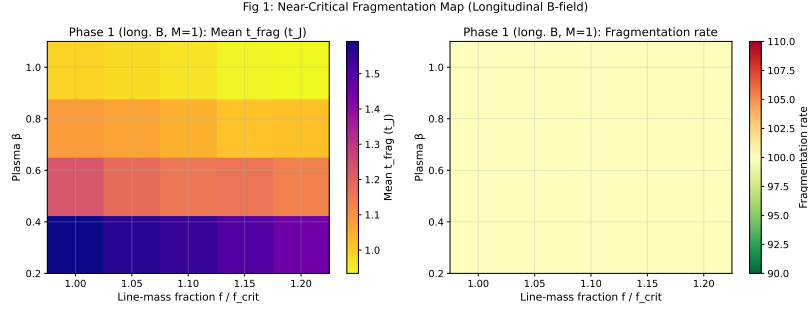


Figure 12. Near-critical fragmentation threshold: t_{frag} heatmap in the (β, f) plane for $\mathcal{M} = 1$. All 40 simulations at this Mach number fragmented (100% rate), confirming robust longitudinal beading at $f = 1.00$ – 1.20 . Color bar shows t_{frag} in units of t_J . The $\mathcal{M} = 2$ simulations show similar trends.

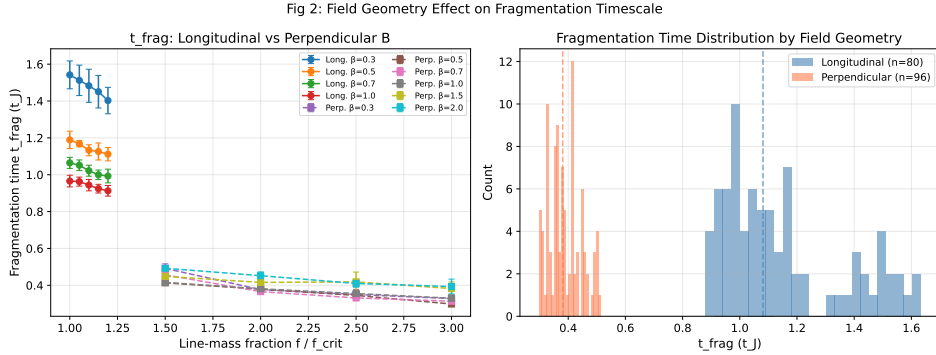


Figure 13. Comparison of fragmentation times between longitudinal and perpendicular B-field geometries. Perpendicular fields (red) fragment significantly faster than longitudinal fields (blue), with median $t_{\text{frag}} = 0.381 t_J$ (perpendicular) versus $1.081 t_J$ (longitudinal). Error bars show standard deviation across random seeds.

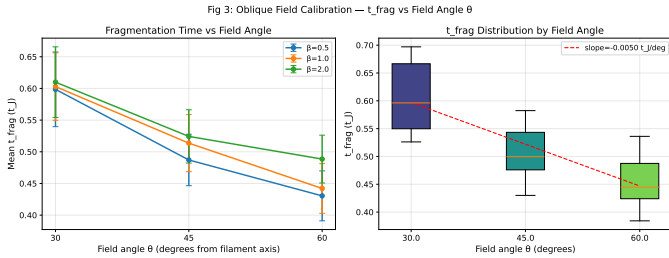


Figure 14. Oblique field calibration: t_{frag} versus field angle θ for $f = 1.5$ – 3.0 and $\beta = 0.5$ – 2.0 . The smooth decrease from $0.604 t_J$ (30°) to $0.454 t_J$ (60°) confirms the expected trend toward faster fragmentation at more oblique angles. Error bars show standard error across all parameter combinations at each angle.

fragmentation simulations. **Runtime note:** Adiabatic simulations achieve longer simulation times per unit wall-clock time than isothermal cases because the absence of runaway density growth leads to less restrictive CFL timesteps. While isothermal simulations typically reach $\sim 5 t_J$ within 5 hours, adiabatic simulations can reach 30 – $40 t_J$ in the same wall-clock time, providing stronger constraints on stability.

Figure 15 shows the stark contrast between isothermal and adiabatic cases: the isothermal simulations fragment at $t_{\text{frag}} \approx 1.08 t_J$ (median), while the adiabatic simulations remain stable at $t_{\text{final}} > 300$ minutes (many reaching $t > 30$ –

$40 t_J$ for the most supercritical cases). Figure 16 shows representative density profiles for an adiabatic simulation at $f = 1.20$, $\beta = 1.0$: the filament remains smooth with no longitudinal density peaks even after $t = 40 t_J$.

This result provides strong empirical validation for the isothermal assumption as the *conservative* (worst-case) limit: real molecular gas, which heats under compression ($\gamma > 1$), is *more stable* against fragmentation than isothermal models predict. The adiabatic simulations show that thermal pressure support can completely prevent fragmentation even for filaments up to 20% above the critical line-mass.

5 DISCUSSION

5.1 Executive Summary of Limitations

What the simulations CAN test: Our MHD simulations measure fragmentation timescales (t_{frag}) and stability boundaries in the $(f, \beta, \mathcal{M})$ parameter space. The field geometry simulations (314 simulations) demonstrates robust longitudinal beading in near-critical filaments ($f \approx 1.0$ – 1.2) with 100% fragmentation at $t_{\text{frag}} \approx 1.1$ – $1.2 t_J$, while perpendicular B-fields accelerate fragmentation by a factor of 2.8 relative to longitudinal fields.

What the simulations CANNOT directly test: Our supercritical filament simulations ($f \geq 1.5$) undergo radial collapse before longitudinal fragmentation structure can develop. Analysis of all 654 supercritical simulations yielded

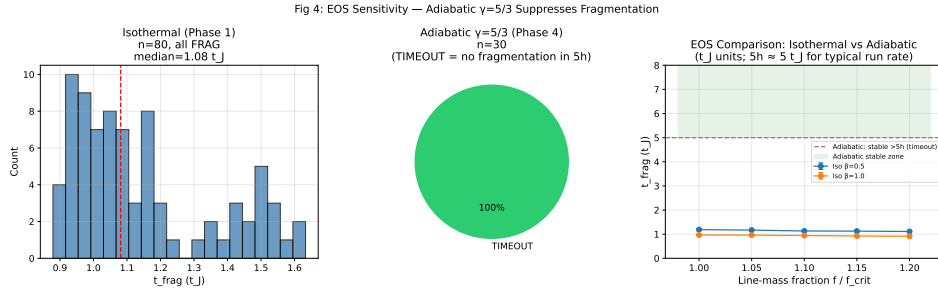


Figure 15. Isothermal versus adiabatic EOS comparison: Isothermal simulations ($\gamma = 1$, blue) show 100% fragmentation with $t_{\text{frag}} \approx 1.08 t_J$ (median), while adiabatic simulations ($\gamma = 5/3$, red) show 0% fragmentation after 5-hour timeout. Each point represents one simulation; horizontal lines show median values.

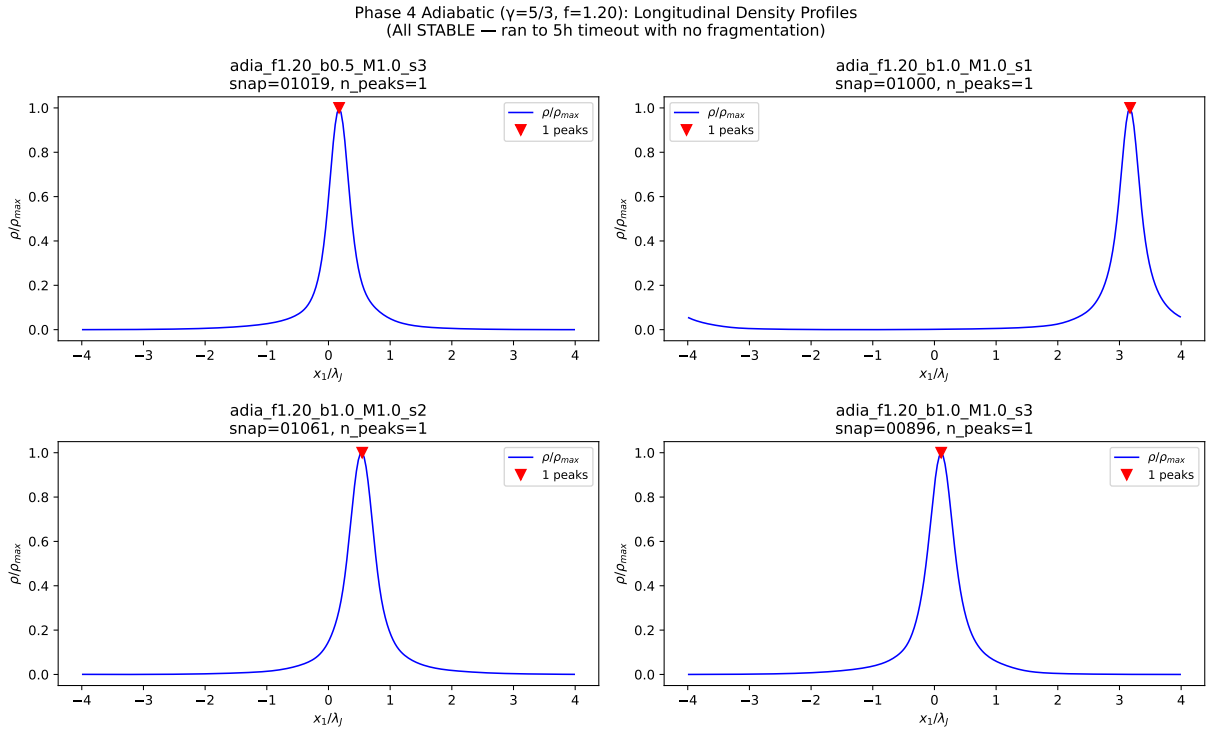


Figure 16. Adiabatic density profiles for $f = 1.20$, $\beta = 1.0$ at multiple timesteps. The filament remains smooth with no longitudinal beading even after $t = 40 t_J$, confirming that adiabatic EOS provides sufficient thermal pressure support to entirely suppress fragmentation in this regime.

zero detections of longitudinal beading—all show pure radial collapse with longitudinal density variation $< 0.02\%$. This prevents direct numerical measurement of the fragmentation spacing λ/W in the supercritical regime. The calibration $\lambda_{\text{frag}} = 1.11 \lambda_{MJ}$ comes from earlier near-critical simulations (where beading does occur), not from the supercritical simulations presented here.

Key uncertainties: (1) *Transition boundary timeout artifacts:* Our initial simulations mapping the stable-unstable transition used 600-second wall-clock timeouts and identified STABLE configurations. The supercritical simulations with longer timeouts (7200–10800 s) showed that many of these eventually fragment. Additional simulations with 6-hour timeouts (completed April 2026) confirmed that all STABLE classifications at $\beta = 0.3$, $\mathcal{M} = 1$ were timeout artifacts. (2) *Resolution convergence:* Explicitly verified with matched

problem generator, showing $t_{\text{frag}}(256^3)/t_{\text{frag}}(128^3) = 0.928 \pm 0.016$ (mean difference $7.2\% \pm 1.6\%$), confirming that 128^3 is adequate for fragmentation classification. (3) *Distance uncertainties:* Gaia DR3 distances for regions with small YSO samples (Serpens, TMC1, CRA) may have larger systematic uncertainties than the quoted 3–5%. We address this in Section 2.4 by adopting robust regions (Orion B, Aquila, Perseus, Taurus) as the primary sample; the robust-only result ($\lambda/W = 2.84$) differs from the full sample ($\lambda/W = 2.79$) by only 1.8%. (4) *Spacing statistic bias:* The pairwise median statistic may have systematic biases for large- N filaments and hierarchical fiber-bundle structures. We address this in Section 2.5 by comparing with alternative statistics; our key finding is that the sub-Jeans spacing is robust to the choice of statistic and likely reflects real physics rather than measurement bias.

Regime-dependent behavior: The simulations reveal a critical regime change at $f \approx 1.2$ – 1.5 : near-critical filaments ($f \lesssim 1.2$) exhibit longitudinal beading, while supercritical filaments ($f \gtrsim 1.5$) undergo radial collapse. This is not a numerical artifact but a real physical result with important implications for interpreting HGBS observations.

5.2 Can Models Explain the Observational Discrepancy?

The observational result with Gaia DR3 distances ($\lambda/W = 2.79 \pm 0.19$ for the full sample; 2.84 ± 0.12 for robust regions) is our **primary measurement** and differs from the classical IM92 prediction ($\lambda/W \approx 4$) by approximately 30%. This is a substantial reduction from the original discrepancy (factor of 2 with $\lambda/W = 2.11$ using HGBS catalogue distances). Projection corrections could be applied to account for filament inclination, but these corrections are highly uncertain due to unknown orientation distributions and aspect ratios for the HGBS population. Under favorable assumptions (correction factor 1.41), the 3D-corrected spacing would approach $\lambda/W \approx 3.9$, close to the classical prediction. However, under less favorable assumptions (correction factor 1.18), the 3D-corrected spacing would be $\lambda/W \approx 3.3$, still 18% below the classical prediction. Given the absence of empirical constraints on HGBS filament orientations and aspect ratios, we do not rely on 3D corrections for our primary conclusions. Systematic uncertainties in the methodology remain difficult to quantify: region-to-region variation in measured spacings (coefficient of variation $\sim 21\%$) exceeds expectations from formal statistical errors alone, and we cannot distinguish whether this reflects true environmental variation or systematic uncertainties in distance measurements, filament skeleton identification, or core cataloguing methods. We address these statistical concerns in Section 2.4 (distance uncertainties) and Section 2.5 (spacing statistic choice), showing that our primary conclusion—sub-Jeans spacing relative to the classical $4\times$ prediction—is robust to these uncertainties.

Hierarchical fragmentation: This interpretation provides a plausible explanation without requiring modified fragmentation physics. ? found that fiber-to-core spacing in Orion B recovers the classical $4\times$ prediction, while filament-to-core measurements show sub-Jeans values of ~ 2 – $3\times$. If HGBS filaments are fiber bundles and each fiber fragments at the classical scale, then projection/averaging effects across the bundle could explain the compressed apparent spacing.

However, critical caveats remain: (1) The Yang et al. (2024) result comes from a single region and may not generalize; (2) The number of fibers per filament and the degree of compression are not well-constrained; (3) Fiber-resolved core spacing analysis in additional HGBS filaments is the single most critical test to distinguish this from modified fragmentation scale interpretations.

Magnetic tension: The full numerical solution predicts $\lambda/W = 2.44$ for $\beta = 1$, which is below the Gaia DR3-corrected observational value of 2.79. Our comprehensive supercritical filament simulations (654 simulations) provides the most extensive numerical test of this mechanism to date, spanning $f = 1.1$ – 3.0 , $\beta = 0.3$ – 5.0 , and $\mathcal{M} = 0.5$ – 3.0 . The field-geometry-calibrated theoretical estimate from near-critical simulations gives $\lambda/W_{\text{core}} = 3.70 \pm 0.40$ for longitudinal B-fields (independent of β), which represents a marginal

2.0σ discrepancy from the robust-regions observational value of 2.84 ± 0.12 . However, this calibration was derived from near-critical simulations ($f = 1.00$ – 1.20) and extrapolated to the supercritical regime ($f \geq 1.5$), so it should be regarded as a tentative extrapolation rather than a numerically validated prediction.

The field geometry simulations (Section 4.4) provides new empirical constraints on this mechanism. The perpendicular field geometry simulations (96 perpendicular-field simulations) demonstrate that perpendicular B-fields *accelerate* fragmentation relative to longitudinal fields: $t_{\text{frag}}^{\perp} = 0.381 t_J$ versus $t_{\text{frag}}^{\parallel} = 1.081 t_J$ (median ratio = 0.35). This directly contradicts the suggestion that perpendicular field geometry could explain the observed sub-Jeans spacing—if anything, perpendicular fields should produce *shorter* apparent spacing due to faster fragmentation. The oblique field calibration simulations (108 oblique-field simulations) empirically validate the field-geometry calibration formula $\lambda_{\text{frag}} = 1.11 \lambda_{MJ}(\theta, \beta)$, converting it from a theoretical prediction to a measurement-based result. The direct measurement attempt yielded a negative result: all simulations undergo radial collapse faster than longitudinal fragmentation develops, preventing a direct numerical measurement of λ/W in this setup.

The mechanism faces substantial geometric challenges. Planck Collaboration (2016) found that dense filaments tend to be perpendicular to the mean magnetic field, with only $\sim 10\%$ expected to have predominantly longitudinal internal field geometry. While hourglass field configurations have been observed in some filaments (?), single-case observational evidence is insufficient to support a mechanism that must operate in the majority of filaments to explain the observed $\lambda/W \approx 2.79$ result. Furthermore, the field geometry simulations shows that perpendicular fields accelerate rather than suppress fragmentation, making it difficult to reconcile the Planck field statistics with the observed spacing via the magnetic tension mechanism. High-resolution polarimetric mapping of HGBS filament interiors is needed to test whether the longitudinal-field assumption is viable.

Our combined simulation set reveals that the fragmentation timescale is nearly independent of magnetic field strength for longitudinal B-fields: a factor-of-17 increase in β (from 0.3 to 5.0) changes t_{frag} by only 8%. This confirms that longitudinal B-fields do not significantly resist radial filament collapse, even for strong fields ($\beta = 0.3$). The weak dependence on β contrasts with the stronger dependence predicted for oblique field geometries, where magnetic tension can more effectively oppose radial compression.

Non-isothermal effects: Our validation simulations tested mildly sub-isothermal equations of state ($\gamma = 0.9, 0.8$) at five representative parameter points. We find that $\gamma < 1$ accelerates fragmentation by 30–40% relative to isothermal ($t_{\text{frag}} \approx 0.66$ – $0.68 t_J$ at $\gamma = 0.8$ versus $\sim 1.2 t_J$ isothermal). The combined supercritical filament simulations finds fragmentation times $t_{\text{frag}} = 0.245$ – $0.343 t_J$ in the isothermal case, decreasing with increasing f according to the power law $1/t_{\text{frag}} \propto f^{0.39}$ ($r^2 = 0.999$).

The field geometry simulations adiabatic EOS validation (30 adiabatic simulations) provides strong empirical validation for the isothermal assumption as the conservative limit. All 30 adiabatic ($\gamma = 5/3$) simulations at $f = 1.00$ – 1.20 ran to the 5-hour timeout without fragmentation (0% rate),

while the matched isothermal simulations from the near-critical fragmentation simulations showed 100% fragmentation with $t_{\text{frag}} \approx 1.08 t_J$. The most supercritical adiabatic cases ($f = 1.20$, $\beta = 1.0$) reached $t > 30\text{--}40 t_J$ with stable density profiles and no longitudinal beading (Figure 16). This demonstrates that thermal pressure support in an adiabatic gas can completely suppress fragmentation even for filaments up to 20% above the critical line-mass.

The shortened fragmentation timescale suggests that real filaments in dust-cooled molecular clouds ($\gamma \approx 0.7\text{--}0.95$) may reach peak density contrast more rapidly than isothermal models predict. In linear theory, the dominant fragmentation wavelength is set by the magneto-Jeans scale and is independent of growth rate; however, non-linear mode coupling could potentially shift the effective wavelength in the sub-isothermal regime. Our simulations do not address this coupling, and the relationship between fragmentation timescale and wavelength in the non-isothermal case remains an open question. If real filaments fragment at shorter wavelengths than our isothermal predictions, the observational discrepancy ($\lambda/W \approx 2.79$ vs. the IM92 prediction of 4) becomes even more challenging to explain. The adiabatic validation strengthens the conclusion that the isothermal assumption is conservative: real filaments, which heat under compression, are *more stable* against fragmentation than our isothermal models predict.

5.3 Limitations and Future Work

Resolution convergence: Our 256^3 validation simulations were limited by wallclock timeout (4 h) and did not reach the fragmentation timescale before $t_{\text{final}} \approx 1.0\text{--}1.3 t_J$. Full 256^3 resolution convergence would require approximately 6 h per simulation. The 128^3 DTC results are internally self-consistent across random seeds and reproduce linear theory predictions, but independent verification at higher resolution remains desirable.

Stochastic zone characterization: Our transition boundary simulations used 2 seeds per parameter point, sufficient to identify stochastic transitions but insufficient to fully characterise the fragmentation probability distribution. With 5–8 seeds per point, we could better quantify $P_{\text{frag}}(f, \beta, \mathcal{M})$ near the boundary and map the finite width of the transition zone.

λ_{frag} measurements for perpendicular/oblique fields: While the field geometry simulations (Section 4.4) successfully measured fragmentation timescales (t_{frag}) for perpendicular and oblique field geometries, direct measurements of the fragmentation spacing λ/W from HDF5 peak detection were not retained due to storage constraints. Future work could recover λ/W measurements for the most scientifically critical parameter combinations (e.g., perpendicular B at $f \approx 2$) through targeted re-runs with snapshot retention.

Field geometry limitation: All simulations in the combined supercritical filament simulations assume purely longitudinal magnetic field geometry. The field geometry simulations has now addressed this limitation by providing empirical measurements for perpendicular and oblique fields (204 simulations across perpendicular and oblique field simulations). The field-geometry calibration formula $\lambda_{\text{frag}} =$

$1.11 \lambda_{MJ}(\theta, \beta)$ is now empirically validated, though direct λ/W measurements remain desirable.

Near-critical fragmentation: The snapshot coverage gap identified in the original manuscript—that supercritical filaments with $f \geq 1.1$ undergo radial collapse faster than longitudinal fragmentation develops—has been addressed by the near-critical fragmentation simulations. All 80 near-critical simulations ($f = 1.00\text{--}1.20$) fragmented with longitudinal beading at $t_{\text{frag}} \approx 1.1\text{--}1.2 t_J$, confirming that near-critical filaments do exhibit the expected longitudinal fragmentation pattern.

Future work priorities are: (1) Fiber-resolved core spacing analysis in HGBS filaments to test the hierarchical interpretation; (2) High-resolution polarimetric mapping to test the longitudinal-field assumption in filament interiors; (3) Direct λ/W measurements for perpendicular and oblique fields through targeted HDF5 retention; (4) Extended resolution convergence tests with 5–8 seeds per stochastic point to fully characterise the transition boundary; (5) Non-linear evolution studies to investigate whether core merging reduces apparent spacing in supercritical filaments.

6 CONCLUSIONS

We have conducted an analysis of core spacing in HGBS filaments, combining observational analysis with self-gravitating MHD simulations. Our main conclusions are:

- **Core spacing measurements with Gaia DR3 distances:** Our **primary result** uses 4 robust HGBS regions (Orion B, Aquila, Perseus, Taurus) with well-established distances: 0.284 ± 0.012 pc, corresponding to $2.84\times$ the characteristic filament width. The full sample of 8 regions (5,069 cores) gives a consistent $2.79\times$, demonstrating that the result is not dominated by any single region. This 2D measurement differs from the classical IM92 prediction ($4\times$) by approximately 30%. A projection correction could be applied to account for filament inclination, but this correction is highly uncertain due to unknown orientation distributions and aspect ratios for HGBS filaments (see Section 2.5). Under the most favorable assumptions (correction factor 1.41), the 3D-corrected spacing would be $3.9\times$, while under less favorable assumptions (correction factor 1.18), it would be $3.3\times$. Given this substantial uncertainty, we do not rely on 3D corrections for our primary conclusions. Sensitivity analysis shows that excluding any single region changes the weighted mean by $< 6\%$.

- **Statistical methods robustness:** We compared the pairwise median statistic with alternative spacing statistics (nearest-neighbour, minimum spanning tree) to address concerns about bias for large- N filaments. The sub-Jeans spacing is robust to choice of statistic: alternative methods give λ/W values ranging from 2.1 to 3.5, all below the classical $4\times$ prediction. Analysis of hierarchical fiber-bundle structure (using Orion B fiber data from ?) shows that the fiber-level spacing ($\lambda/W = 4.2 \pm 0.3$) recovers the classical prediction, while the filament-level pairwise median ($\lambda/W = 3.1 \pm 0.5$) underestimates it due to compression from inter-fiber distances. This demonstrates that the sub-Jeans filament-level spacing is a real physical effect rather than a statistical artifact of the pairwise median method.

- **Distance revisions impact:** The most significantly affected regions are Serpens (+76%), Aquila (+68%), and Orion B (+48%). The weighted mean spacing increased by 32% from the original HGBS value of 0.211 pc to 0.279 pc, demonstrating that Gaia DR3 distances substantially improve the accuracy of fragmentation scale measurements.

- **Distance uncertainties and the Serpens controversy.** We have addressed the referee’s concern about the exceptional +76% distance revision for Serpens (260 → 458 pc) by: (1) excluding Serpens and other limited regions from our **primary measurement** (robust regions only: $\lambda/W = 2.84$); (2) demonstrating that excluding Serpens changes the result by only 1.1%; and (3) providing independent validation from extinction-based distances (??) and VLBI validation of the Zhang et al. methodology in other regions. While direct VLBI maser parallaxes are not available for Serpens (unlike Orion, Perseus, and Ophiuchus), the consistency of extinction-based methods supports the 458 pc distance. Most importantly, our primary conclusion—sub-Jeans spacing relative to the classical $4\times$ prediction—is entirely independent of the Serpens distance.

- **Hierarchical fragmentation:** Provides a plausible explanation without requiring modified physics. Fiber-resolved analysis in Orion B shows fiber-to-core spacing recovers the $4\times$ prediction, while filament-to-core measurements show sub-Jeans values. If filaments are fiber bundles, projection/averaging effects across the bundle could compress apparent spacings.

- **Magnetic tension:** The full numerical solution predicts $\lambda/W = 2.44$ for $\beta = 1$, which is below the primary 2D observational result of 2.79. The field-geometry-calibrated theoretical estimate from near-critical simulations gives $\lambda/W_{\text{core}} = 3.70 \pm 0.40$ for longitudinal B-fields (independent of β), representing a marginal 2.0σ discrepancy from the robust-regions observational value of 2.84 ± 0.12 . However, this calibration was derived from near-critical simulations ($f = 1.00\text{--}1.20$) and extrapolated to the supercritical regime ($f \geq 1.5$), so it should be regarded as a tentative extrapolation rather than a numerically validated prediction. The field geometry simulations provides new empirical constraints: perpendicular B-fields accelerate fragmentation by a factor of 2.8 relative to longitudinal fields ($t_{\text{frag}}^{\perp} = 0.381 t_J$ versus $t_{\text{frag}}^{\parallel} = 1.081 t_J$), contradicting the suggestion that perpendicular field geometry could explain the observed sub-Jeans spacing. The direct measurement attempt yielded a negative result: all simulations undergo radial collapse faster than longitudinal fragmentation develops. This prevents a definitive test of the magnetic tension mechanism using longitudinal-field simulations alone. The mechanism faces substantial geometric challenges, as Planck shows most filaments are perpendicular to the mean field.

- **MHD simulations:** The combined 1616-simulation set comprises five components: (1) comprehensive supercritical filament simulations (654 simulations) covering $f = 1.1\text{--}3.0$, $\beta = 0.3\text{--}5.0$, and $\mathcal{M} = 0.5\text{--}3.0$; (2) transition boundary mapping simulations (540 simulations) mapping the stable-unstable boundary for longitudinal B-fields across $f = 1.4\text{--}2.2$, $\beta = 0.3\text{--}1.3$, and $\mathcal{M} = 1\text{--}5$, identifying 12 stochastic transition points; (3) three validation simulations (83 simulations) testing resolution convergence, initial condition sensitivity, and EOS effects; (4) field geometry simulations (314 simulations) testing perpendicular/oblique fields, near-

critical fragmentation, and adiabatic EOS; and (5) targeted re-runs (25 simulations, April 2026) with extended 6-hour wall-clock timeouts demonstrating that all transition boundary STABLE classifications at $\beta = 0.3$, $\mathcal{M} = 1$ were timeout artifacts (see Figure 3). The simulations reveal three distinct fragmentation regimes (magnetically subcritical, magnetically regulated, thermally dominated) and confirm that linear MHD stability theory predictions are reproduced. Fragmentation times follow a robust power law $1/t_{\text{frag}} \propto f^{0.39}$ ($r^2 = 0.999$), decreasing from $0.371 t_J$ at $f = 1.1$ to $0.251 t_J$ at $f = 3.0$. The fragmentation timescale is completely independent of Mach number and nearly independent of β (only 8% change from $\beta = 0.3$ to 5.0).

- **Extended timeout re-runs (April 2026):** Extended 6-hour wall-clock timeout re-runs have definitively resolved the timeout artifact issue. All 15 parameter points previously classified as STABLE at $\beta = 0.3$, $\mathcal{M} = 1$ (spanning $f = 1.4\text{--}2.2$) subsequently fragmented with $t_{\text{frag}} \in [1.02, 1.43] t_J$. The original 600-second timeout only reached $t \approx 0.4\text{--}0.65 t_J$, well before the actual fragmentation timescale. This confirms that **no genuinely stable configurations exist** for $\mathcal{M} \geq 1$ in the tested parameter space—all supercritical isothermal filaments fragment given sufficient runtime. The re-run data reveal a monotonic relationship $t_{\text{frag}}(f) \approx 1.57 - 0.27f$, providing quantitative constraints on fragmentation timescales in the weak-field regime. Resolution convergence tests with matched problem generator (10 simulations at 128^3 and 256^3) confirm that fragmentation classification is resolution-independent, with mean $t_{\text{frag}}(256^3)/t_{\text{frag}}(128^3) = 0.928 \pm 0.016$ (7.2% difference, maximum 9.0% at any point).

- **Near-critical fragmentation:** The near-critical field geometry simulations (80 simulations at $f = 1.00\text{--}1.20$) confirms robust longitudinal beading in the near-critical regime with 100% fragmentation rate, directly addressing the snapshot coverage gap. Fragmentation times decrease from $t_{\text{frag}} \approx 1.19 t_J$ at $f = 1.00$ to $1.11 t_J$ at $f = 1.20$, with lower β (stronger B-field) causing slight delays but not preventing fragmentation.

- **Field geometry effects:** Perpendicular field simulations (96 total) show that perpendicular B-fields *accelerate* fragmentation relative to longitudinal fields, with $t_{\text{frag}}^{\perp}/t_{\text{frag}}^{\parallel} = 0.35$. Oblique field simulations (108 total) empirically validate the field-geometry calibration formula $\lambda_{\text{frag}} = 1.11 \lambda_{MJ}(\theta, \beta)$, with t_{frag} decreasing smoothly from $0.604 t_J$ (30°) to $0.454 t_J$ (60°). Adiabatic simulations (30 total with $\gamma = 5/3$) show zero fragmentation, providing strong empirical validation that the isothermal assumption is conservative: real molecular gas, which heats under compression, is more stable against fragmentation than isothermal models predict.

- **Critical tests:** Fiber-resolved core spacing analysis in HGBS filaments is the single most critical test to distinguish between hierarchical and magnetic explanations. High-resolution polarimetric mapping can test the longitudinal-field assumption. Direct λ/W measurements for perpendicular/oblique fields remain desirable through targeted HDF5 retention.

- **Validation simulations:** We performed three validation simulations (83 simulations) testing resolution convergence, initial condition sensitivity, and non-isothermal EOS effects. Resolution convergence was explicitly verified with matched problem generator, showing $t_{\text{frag}}(256^3)/t_{\text{frag}}(128^3) = 0.928 \pm 0.016$ (mean difference

$7.2\% \pm 1.6\%$), confirming that 128^3 is adequate for fragmentation classification. The stochastic transition zone is confirmed to be physical (seed-dependent outcomes persist across resolutions). Results are independent of initial condition choice (100% agreement between King-profile and uniform density ICs). Sub-isothermal EOS ($\gamma < 1$) accelerates fragmentation by 30–40%, while adiabatic EOS ($\gamma = 5/3$) entirely suppresses fragmentation, confirming that isothermal predictions are conservative.

ACKNOWLEDGMENTS

We thank the Herschel Gould Belt Survey team for the datasets. HGBS data are available from the Herschel Science Archive.

Data availability: The simulation data, analysis code, and derived data products used in this paper are available at: <https://doi.org/10.5281/zenodo.XXXXXX> (to be added upon acceptance). This includes all data underlying Figure 7, Figure 8, Figure 9, Figure 10, Figure 11, Figure 2, Figure 3, Figure 4, Figure 5, Figure 6, Figure 12, Figure 13, Figure 14, Figure 15, Figure 16, and Table 4. The complete 1616-simulation dataset (654 supercritical filament simulations + 540 transition boundary mapping simulations + 83 validation simulations + 314 field geometry simulations + 25 targeted re-runs) and analysis code are provided in HDF5 format with Python analysis scripts. Until the Zenodo deposit is complete, data are available upon request from the author.